

STAT 516 Lec 02

Review of simple linear regression

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2025-01-28

Hemoglobin versus RBC count example

These data are a subset of the dataset Marcinkevičs et al. (2023)

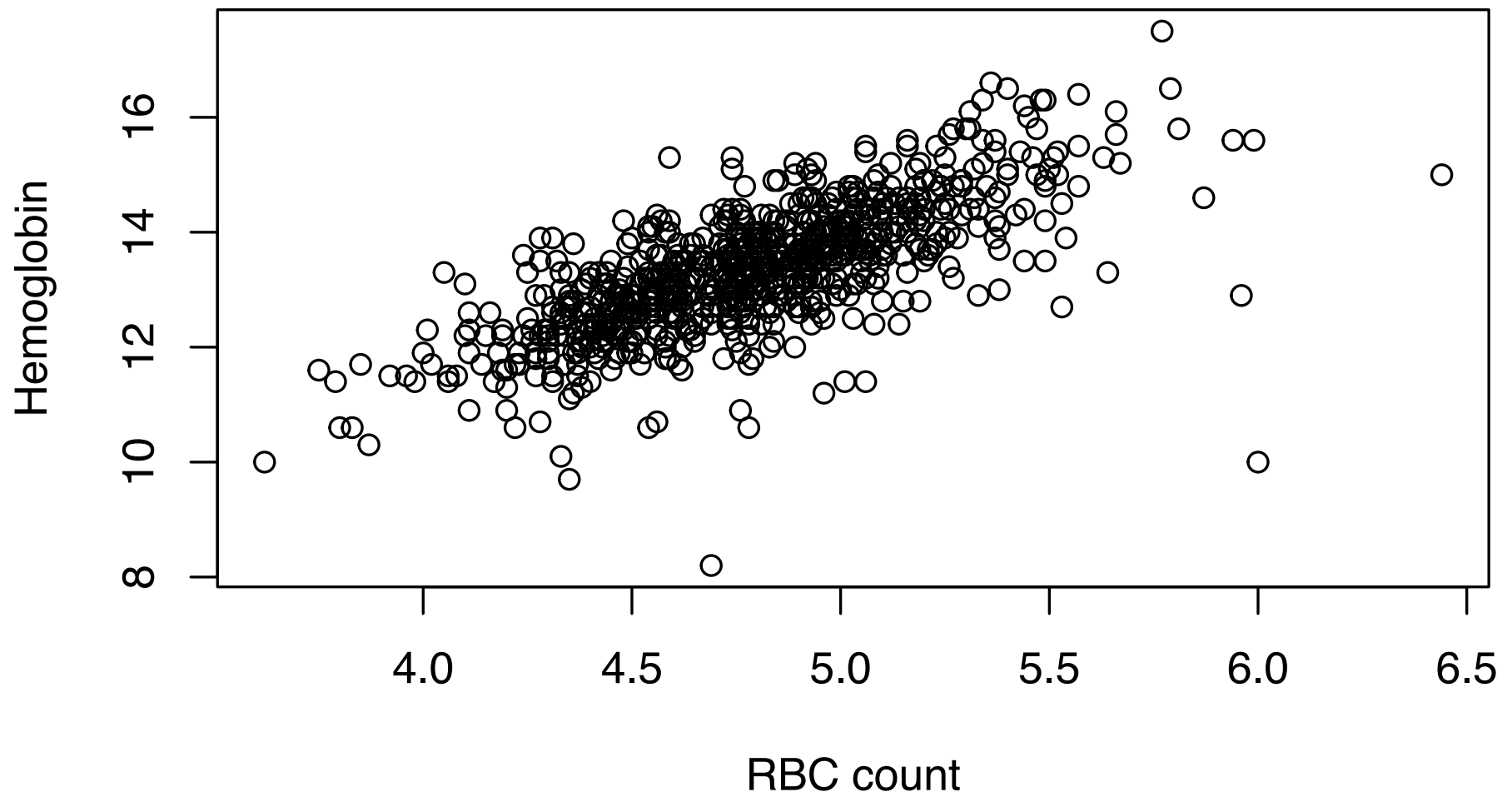
Some outliers and missing values were removed.

```
link <- url("https://people.stat.sc.edu/gregorkb/data/hrbc.csv")
data <- read.csv(link)
head(data)
```

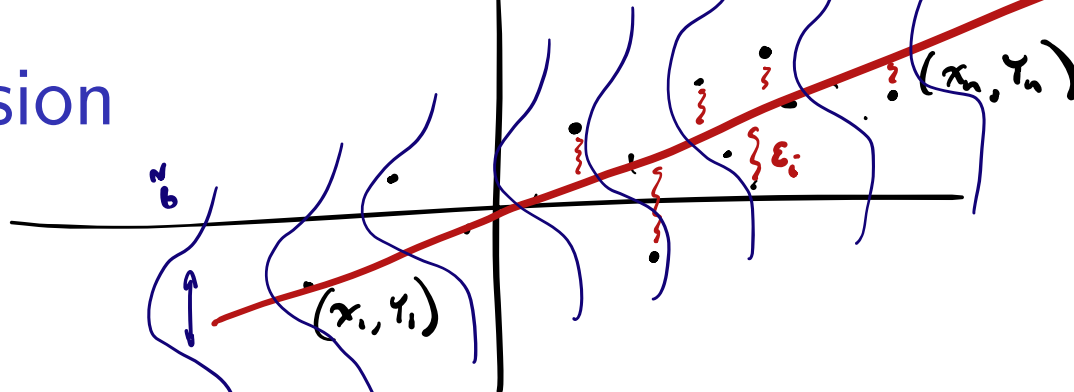
	hem	rbc	sex	age		diag
1	14.8	5.27	female	12.68		appendicitis
2	15.7	5.26	male	14.10	no	appendicitis
3	11.4	3.98	female	14.14	no	appendicitis
4	13.6	4.64	female	16.37	no	appendicitis
5	12.6	4.44	female	11.08		appendicitis
6	12.5	4.96	male	11.05	no	appendicitis

Hemoglobin level vs red blood cell count for $n = 762$ children.

```
plot(data$hem ~ data$rbc,  
      ylab = "Hemoglobin",  
      xlab = "RBC count")
```



Simple linear regression



For $(x_1, Y_1), \dots, (x_n, Y_n)$, the simple linear regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n$$

"epsilon"

"noise", "error term"

where

- ▶ $x_1, \dots, x_n$ are the covariate or predictor values.
- ▶ $Y_1, \dots, Y_n$ are the response values.
- ▶ β_0 and β_1 are the intercept and slope parameters, respectively.
- ▶ $\epsilon_1, \dots, \epsilon_n$ are independent $\text{Normal}(0, \sigma^2)$ error terms.
- ▶ σ^2 is the error term variance.

Goals in simple linear regression

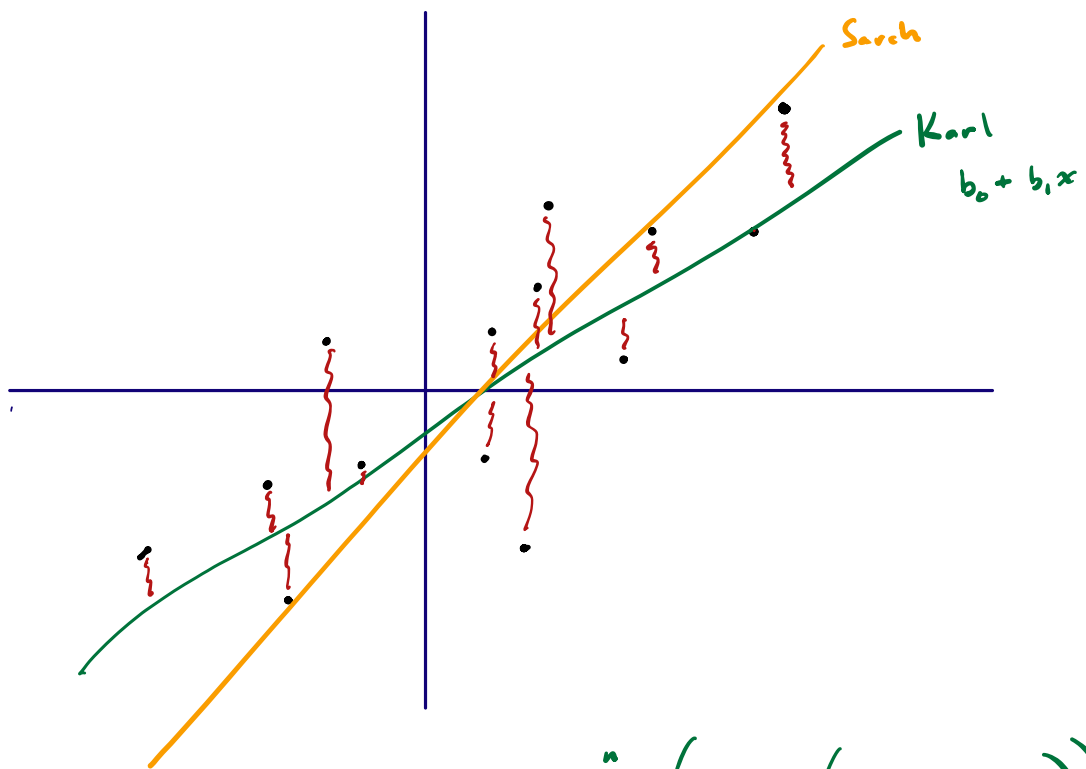
We will learn how to:

1. Estimate the intercept and slope parameters β_0 and β_1 .
2. Estimate the error term variance σ^2 .
3. Perform inference on β_1 .
4. Build a confidence interval for $\beta_0 + \beta_1 x_{\text{new}}$ at any x_{new} .
5. Build a prediction interval for Y at any x_{new} .
6. Decompose the variation in Y into sums of squares.
7. Check whether the model assumptions are satisfied.
8. Identify outliers and understand their effects.

EXTRAS:

1) Two-sample t-test via SLR

2) Do we really need Normal error terms?



$$\sum_{i=1}^n \left(y_i - (b_0 + b_1 x_i) \right)^2$$

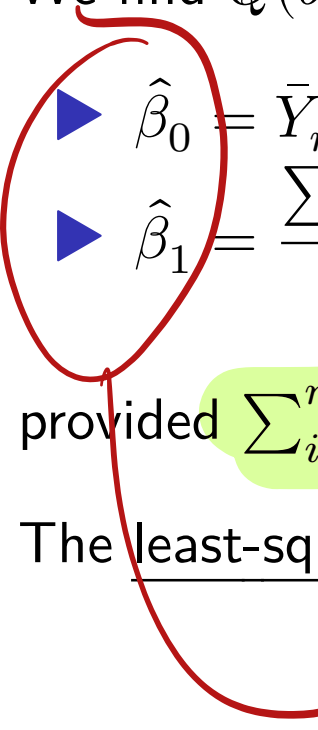
Least-squares estimation of slope and intercept

The squared error criterion given by the sum

$$Q(b_0, b_1) = \sum_{i=1}^n (Y_i - (b_0 + b_1 x_i))^2$$

of squared vertical distances of Y_i from the line $y = b_0 + b_1 x$.

We find $Q(b_0, b_1)$ is minimized at $(b_0, b_1) = (\hat{\beta}_0, \hat{\beta}_1)$, where


$$\begin{aligned} \hat{\beta}_0 &= \bar{Y}_n - \hat{\beta}_1 \bar{x}_n \quad \leftarrow \\ \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x}_n)(Y_i - \bar{Y}_n)}{\sum_{i=1}^n (x_i - \bar{x}_n)^2} \quad \checkmark \end{aligned}$$

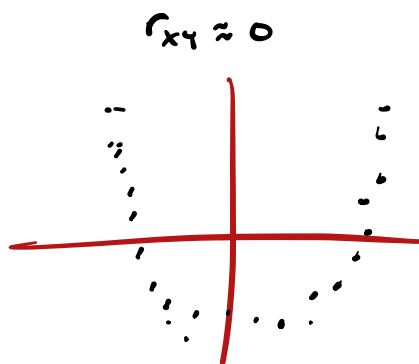
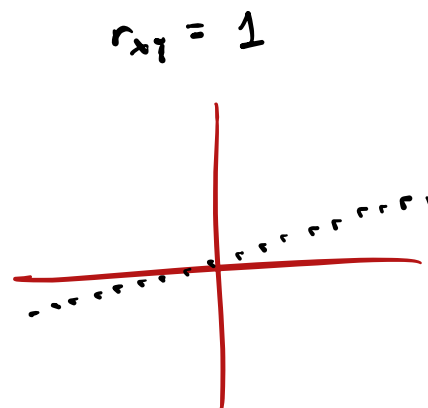
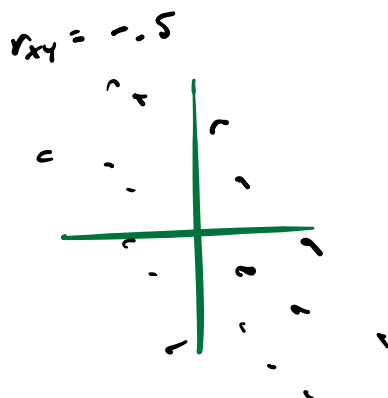
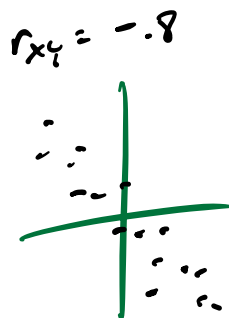
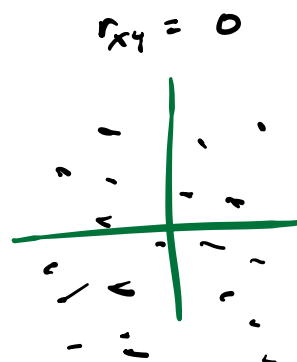
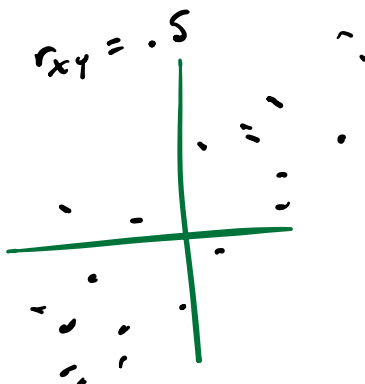
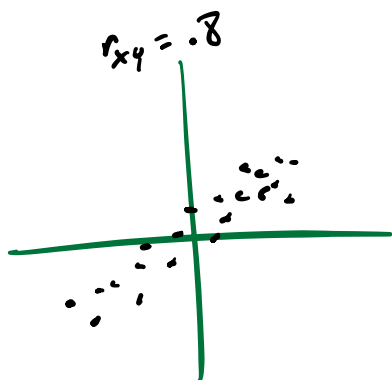
provided $\sum_{i=1}^n (x_i - \bar{x}_n)^2 > 0$.

The least-squares line or fitted line is the line $y = \hat{\beta}_0 + \hat{\beta}_1 x$.

"least-squares estimators"

Pearson's Correlation Coef.

"in"
↓
 $r_{xy} \in [-1, 1]$



Pearson's correlation coefficient

Given $(x_1, Y_n), \dots, (x_n, Y_n)$, the quantity

$$r_{xY} = \frac{\sum_{i=1}^n (x_i - \bar{x}_n)(Y_i - \bar{Y}_n)}{\sqrt{\sum_{i=1}^n (x_i - \bar{x}_n)^2 \sum_{i=1}^n (Y_i - \bar{Y}_n)^2}}$$

is called Pearson's correlation coefficient.

- ▶ Describes strength and direction of *linear* relationships.
- ▶ Must satisfy $r_{xY} \in [-1, 1]$.
- ▶ Values close to zero indicate a weak linear relationship.
- ▶ Is related to $\hat{\beta}_1$ by

$$\hat{\beta}_1 = r_{xY} \frac{S_Y}{S_x}$$

Handwritten annotations: S_Y is labeled $sd(Y)$ and S_x is labeled $sd(x)$.

where S_Y and S_x are the sample std devs of the Y and x values.

$$\text{cor}(x, Y) = \text{cor}(Y, x)$$

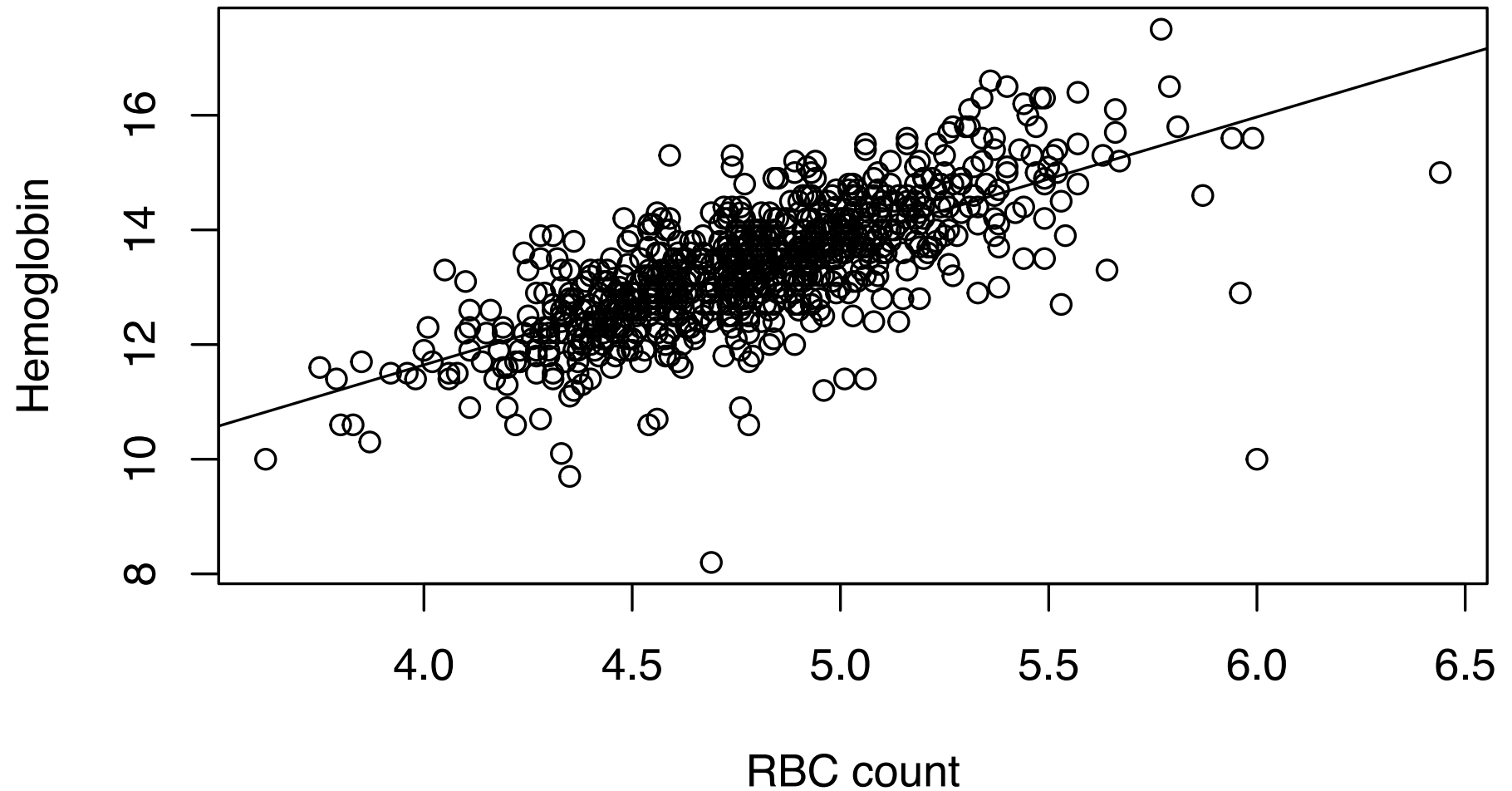
Hemoglobin versus RBC count example (cont)

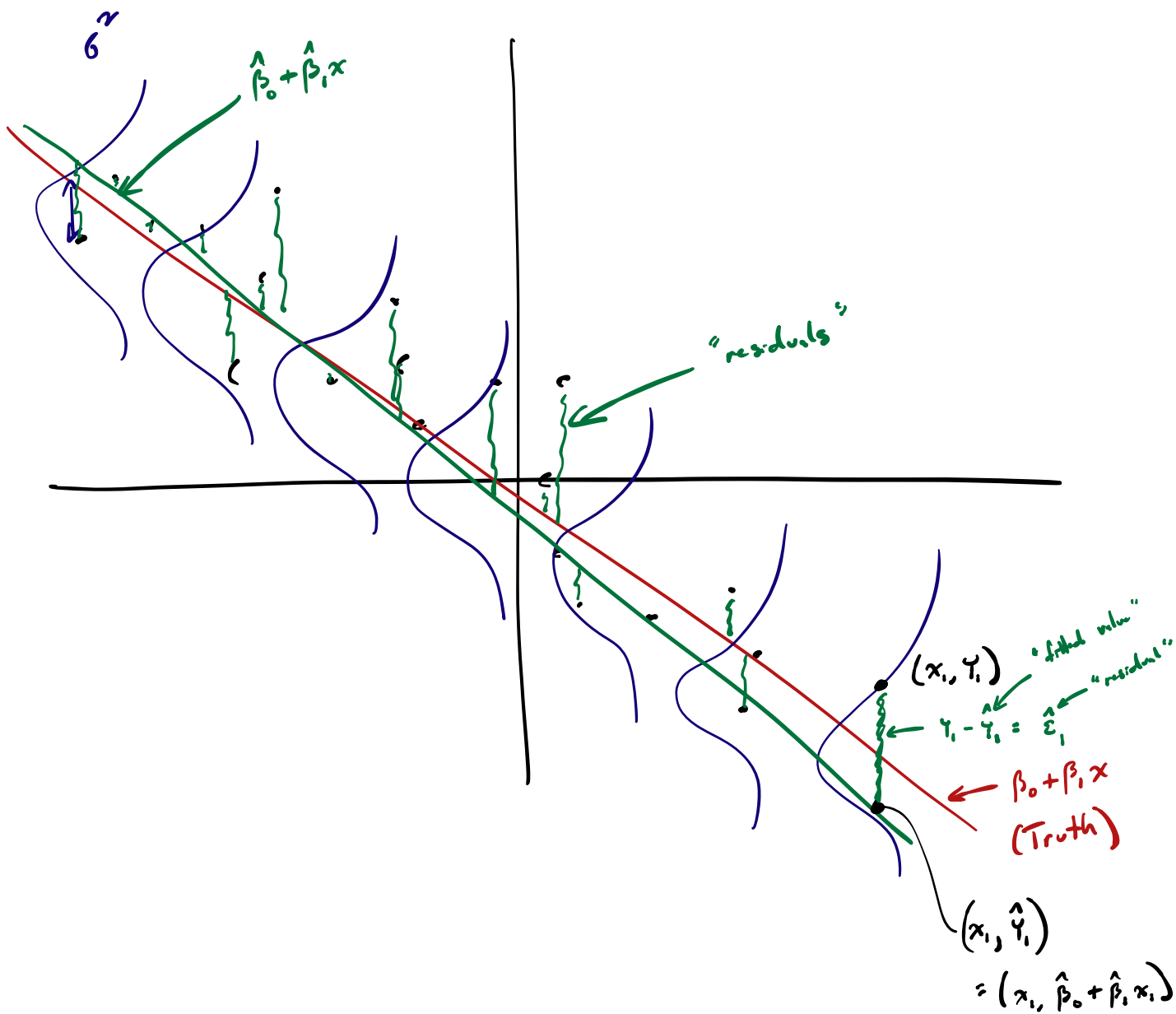
Find the least-squares line on the hemoglobin data.

```
Y <- data$hem
x <- data$rbc
n <- length(Y)
xbar <- mean(x)
Ybar <- mean(Y)

rxY <- cor(x,Y) # Pearson's correlation coefficient
b1hat <- rxY * sd(Y)/sd(x)
b0hat <- Ybar - b1hat * xbar
```

```
plot(data$hem ~ data$rbc, ylab = "Hemoglobin", xlab = "RBC count")  
abline(b0hat, b1hat)
```





Estimating the error term variance

After obtaining $\hat{\beta}_0$ and $\hat{\beta}_1$, define the

- ▶ fitted values as $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ ← Estimated line at the observed value of x .
- ▶ residuals as $\hat{\varepsilon}_i = Y_i - \hat{Y}_i$

for $i = 1, \dots, n$.

Then an unbiased estimator of σ^2 is given by

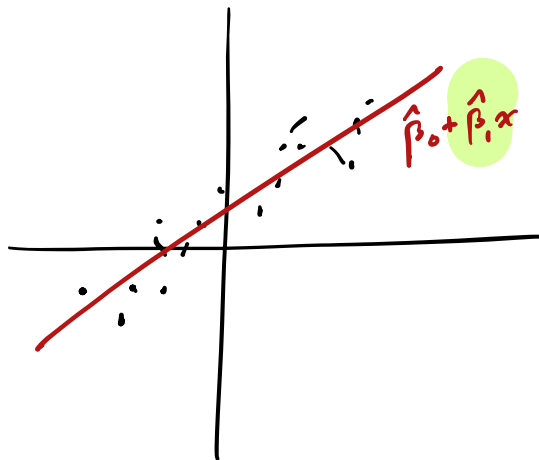
$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{\varepsilon}_i^2.$$

$n-2$ because we had n data points and estimated 2 things already.

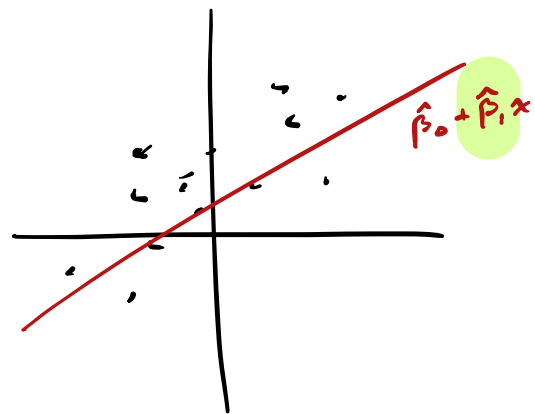
$$= \frac{1}{n-2} \sum_{i=1}^n \left(\hat{\varepsilon}_i - \overline{\hat{\varepsilon}} \right)^2$$

mean of $\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_n$ is equal to zero

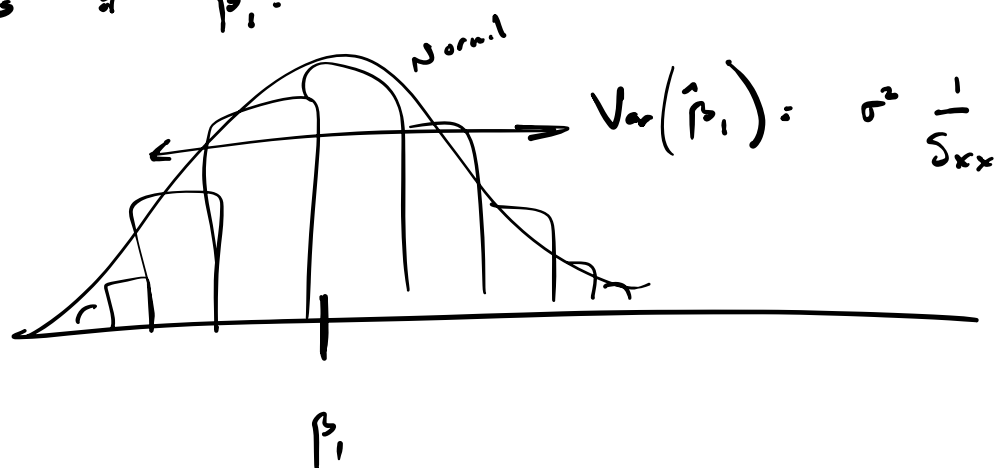
Collect data many times, compute LS estimator $\hat{\beta}_0, \hat{\beta}_1$



...



Many values of $\hat{\beta}_1$:



$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

Confidence interval for the slope parameter

Assume $\varepsilon_1, \dots, \varepsilon_n \stackrel{\text{ind}}{\sim} \text{Normal}(0, \sigma^2)$ and set $S_{xx} = \sum_{i=1}^n (x_i - \bar{x}_n)^2$.

- ▶ The slope estimator $\hat{\beta}_1$ is distributed as

$$\hat{\beta}_1 \sim \text{Normal}(\beta_1, \sigma^2 / S_{xx}).$$

- ▶ “Studentizing” the above gives

← subtract them, divide by the estimated std. dev.

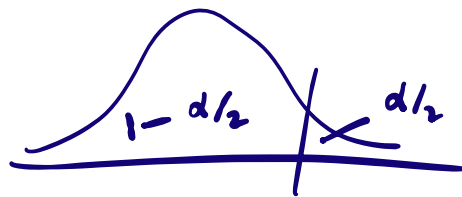
$$\frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma} / \sqrt{S_{xx}}} \sim t_{n-2}.$$

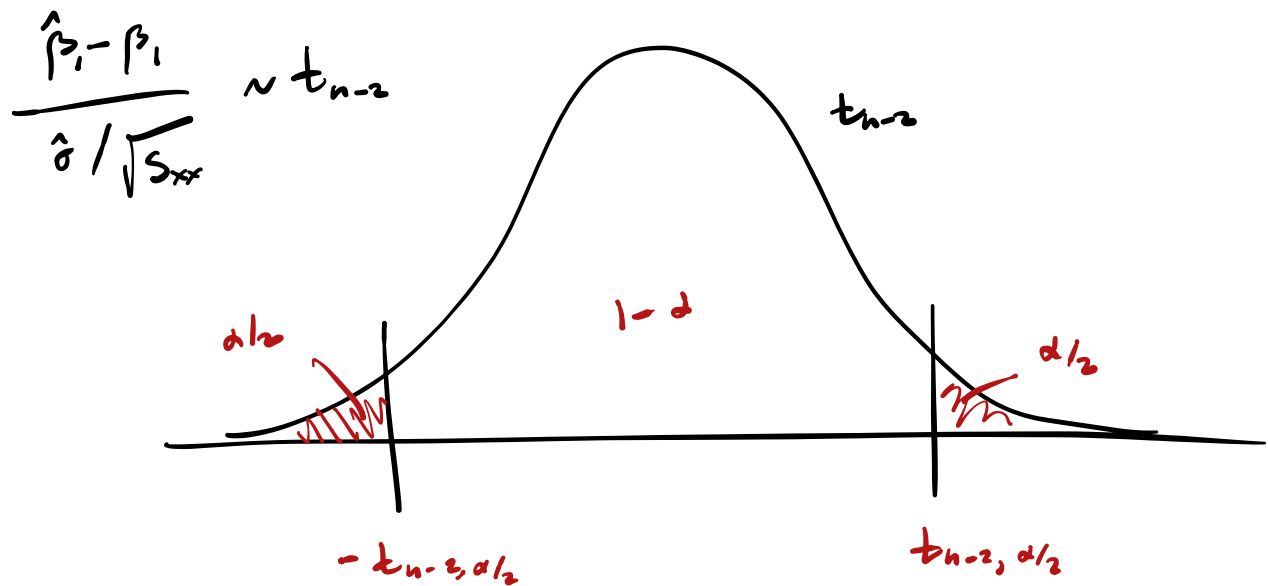
- ▶ So a $(1 - \alpha)100\%$ confidence interval for β_1 is

$t(1 - \alpha/2, n-2)$

$$\hat{\beta}_1 \pm t_{n-2, \alpha/2} \hat{\sigma} / \sqrt{S_{xx}}.$$

Exercise: Justify the CI using the sampling distribution result.





$$\Rightarrow P\left(-t_{n-2, \alpha/2} < \frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma} / \sqrt{S_{xx}}} < t_{n-2, \alpha/2}\right) = 1 - \alpha$$

∴ Rearrange till β_1 is alone in middle

$$P\left(\hat{\beta}_1 - t_{n-2, \alpha/2} \frac{\hat{\sigma}}{\sqrt{S_{xx}}} < \beta_1 < \hat{\beta}_1 + \underline{\underline{t_{n-2, \alpha/2} \frac{\hat{\sigma}}{\sqrt{S_{xx}}}}}\right)$$

$$\Rightarrow (1 - \alpha) \cdot 100\% \text{ C.I. of } \beta_1 \text{ is}$$

$$\hat{\beta}_1 \pm t_{n-2, \alpha/2} \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$$

Hemoglobin versus RBC count example (cont)

Obtain an estimate of the error term variance.

```
Yhat <- b0hat + b1hat * x
ehat <- Y - Yhat
sgsqhat <- sum(ehat^2)/(n-2)
```

We obtain $\hat{\sigma}^2 = 0.626$.

Now construct a 95% confidence interval for β_1 .

```
alpha <- 0.05
Sxx <- sum((x - xbar)^2)
ta2 <- qt(1 - alpha/2, df = n - 2)
lo <- b1hat - ta2 * sqrt(sgsqhat / Sxx)
up <- b1hat + ta2 * sqrt(sgsqhat / Sxx)
```

The 95% CI is (2.011, 2.314).

Note: Can use $\text{lm_out} \leftarrow \text{lm}(Y \sim x)$
 $\text{confint}(\text{lm_out})$

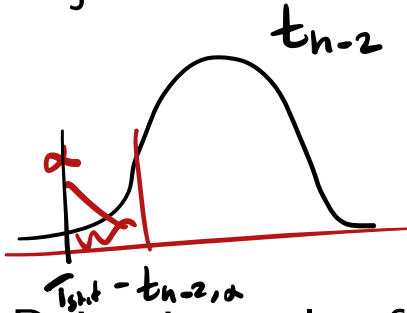
Tests of hypotheses about the slope

If $\beta_1 = 0$, then
X and Y have no
linear relationship

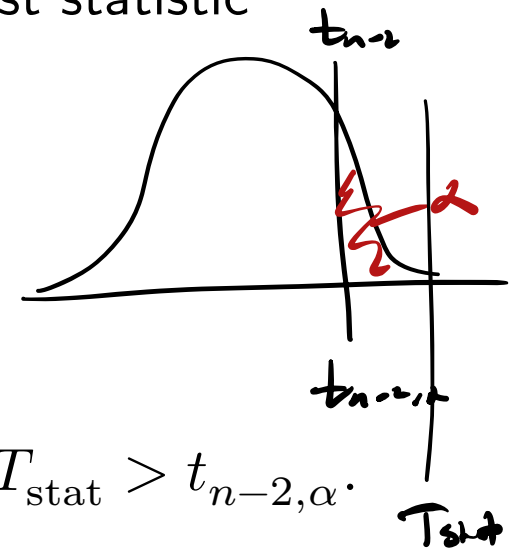
We most often test hypotheses about β_1 of the form

$$\begin{array}{lll} H_0: \beta_1 \geq 0 & \text{or} & H_0: \beta_1 = 0 & \text{or} & H_0: \beta_1 \leq 0 \\ H_1: \beta_1 < 0 & & H_1: \beta_1 \neq 0 & & H_1: \beta_1 > 0. \end{array}$$

Reject or fail to reject H_0 based on the value of the test statistic



$$T_{\text{stat}} = \frac{\hat{\beta}_1 - 0}{\hat{\sigma} / \sqrt{S_{xx}}}$$

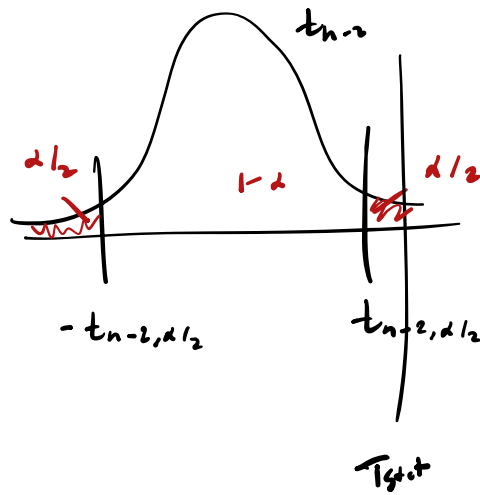


Rejection rules for the above at significance level α are

$$T_{\text{stat}} < -t_{n-2, \alpha} \quad \text{or} \quad |T_{\text{stat}}| > t_{n-2, \alpha/2} \quad \text{or} \quad T_{\text{stat}} > t_{n-2, \alpha}.$$

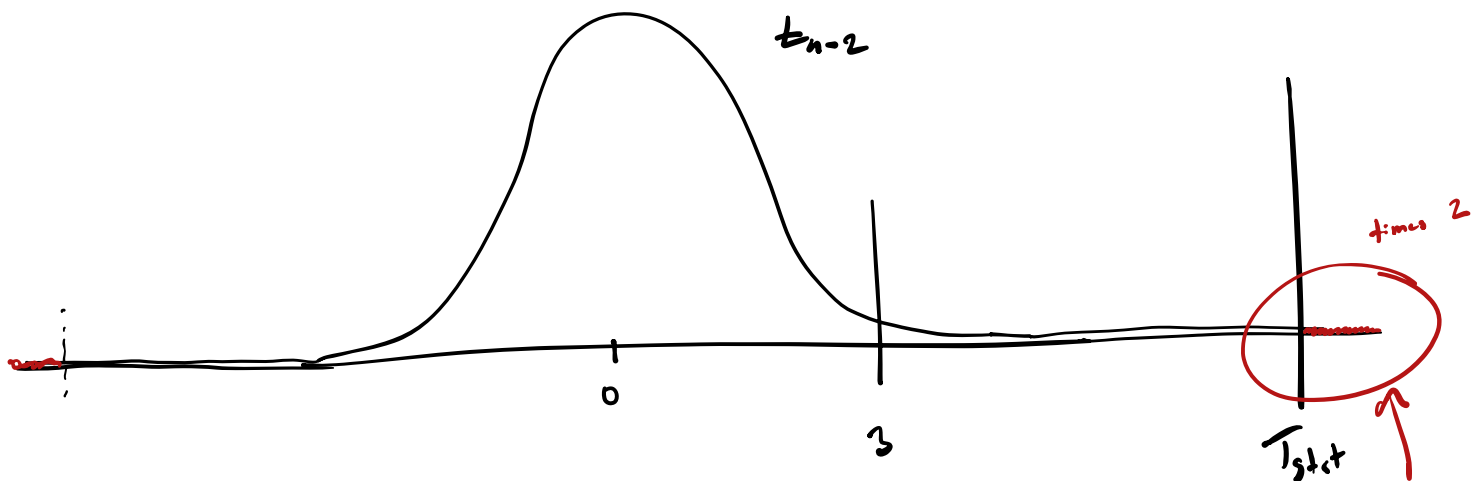
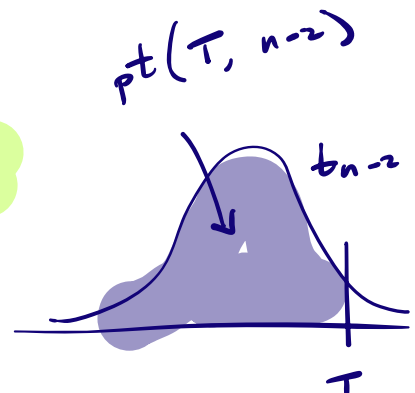
The corresponding p-values are, with $T \sim t_{n-2}$, the probabilities

$$P(T < T_{\text{stat}}) \quad \text{or} \quad 2 \times P(T > |T_{\text{stat}}|) \quad \text{or} \quad P(T > T_{\text{stat}}).$$



T_{stat} $H_0: \beta_1 = 0$ in RBC data
 $H_1: \beta_1 \neq 0$ at $\alpha = 0.05$

$$T_{stat} = \frac{\hat{\beta}_1}{\hat{\sigma} / \sqrt{S_{xx}}} = 28.02$$



$p\text{-val} \approx 0$

$$2 (1 - pt(|T_{stat}|, n-2))$$

Hemoglobin versus RBC count example (cont)

Test the hypotheses $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$ at $\alpha = 0.05$.

```
alpha <- 0.05
Tstat <- b1hat / sqrt(sgsqhat/Sxx)
crit <- qt(1-alpha/2, df = n - 2)
pval <- 2*(1 - pt(abs(Tstat), df = n - 2))
```

We get $T_{\text{stat}} = 28.021$, $t_{n-2, \alpha/2} = 1.963$, and p -value 0; we reject H_0 .

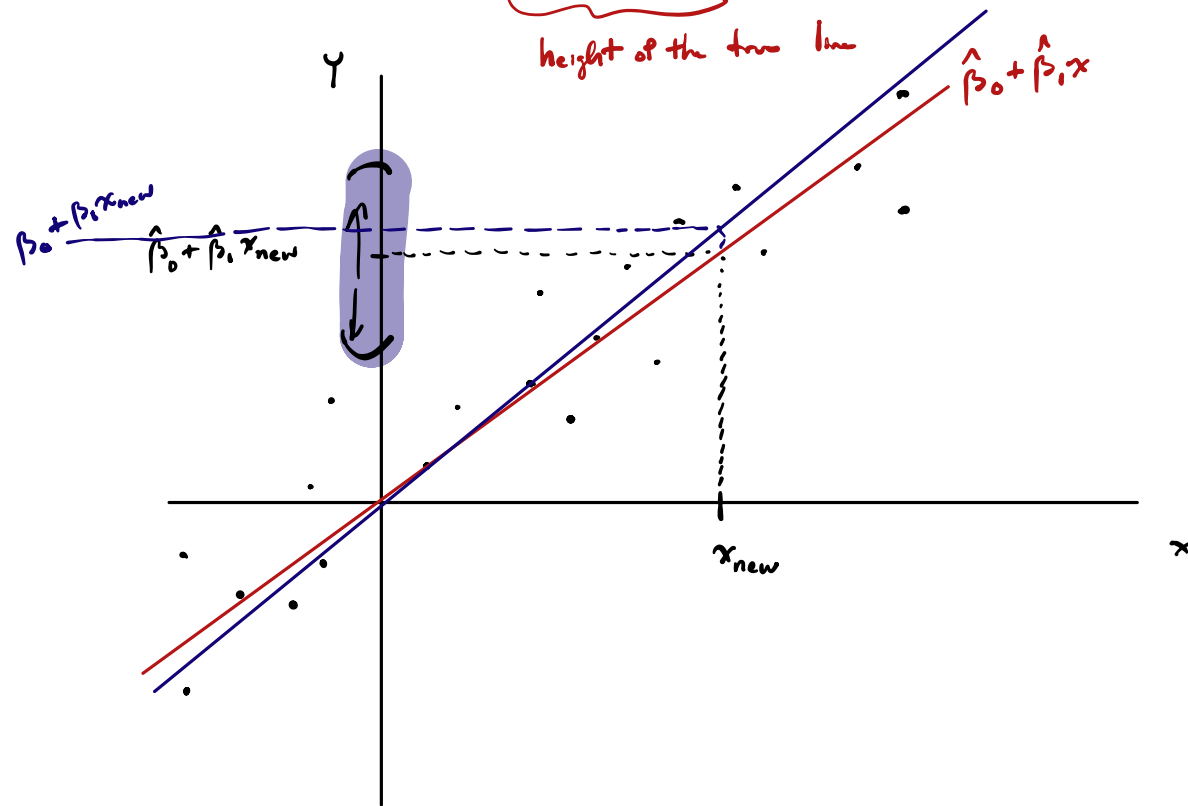
Test the hypotheses $H_0: \beta_1 \leq 2$ vs $H_1: \beta_1 > 2$.

```
Tstat <- (b1hat - 2) / sqrt(sgsqhat/Sxx)
crit <- qt(1-alpha, df = n - 2)
pval <- 1 - pt(Tstat, df = n - 2)
```

We get $T_{\text{stat}} = 2.107$, $t_{n-2, \alpha} = 1.647$, and p -value 0.018; we reject H_0 .

Build C.I. for

$\beta_0 + \beta_1 x_{\text{new}}$ at some x_{new} .
height of the true line



Confidence interval for the height of the line

Assume $\varepsilon_1, \dots, \varepsilon_n \stackrel{\text{ind}}{\sim} \text{Normal}(0, \sigma^2)$. Then:

- ▶ The estimator $\hat{\beta}_0 + \hat{\beta}_1 x$ is distributed as

$$\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \sim \text{Normal} \left(\beta_0 + \beta_1 x_{\text{new}}, \sigma^2 \left[\frac{1}{n} + \frac{(x_{\text{new}} - \bar{x}_n)^2}{S_{xx}} \right] \right).$$

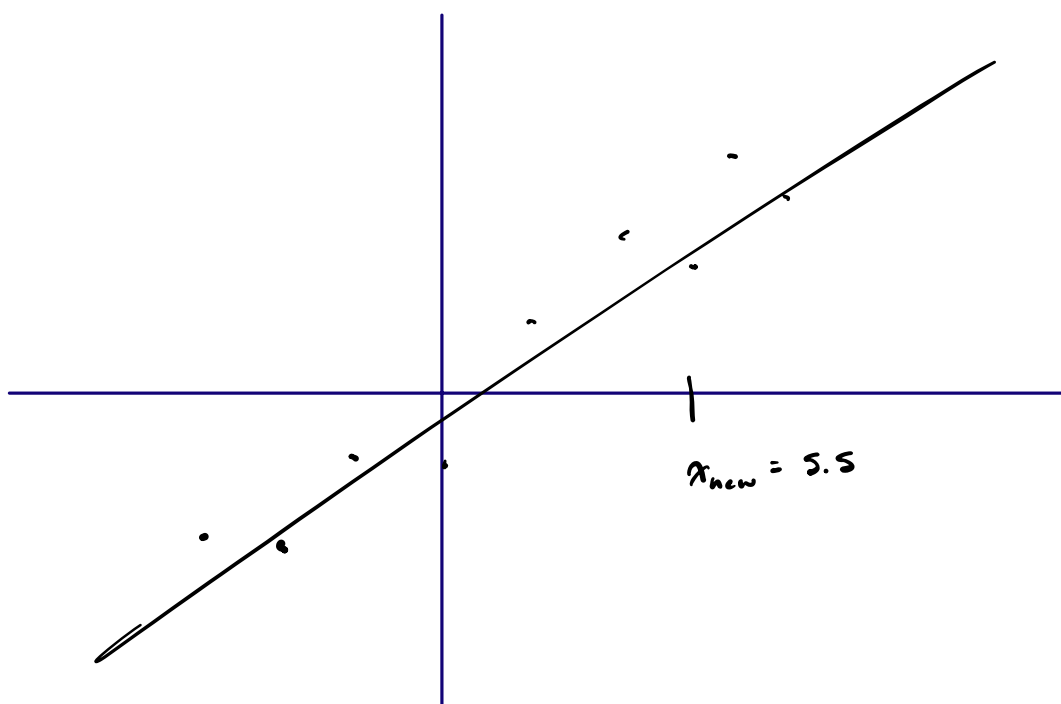
- ▶ “Studentizing” the above gives

$$\frac{\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} - (\beta_0 + \beta_1 x_{\text{new}})}{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{\text{new}} - \bar{x}_n)^2}{S_{xx}}}} \sim t_{n-2}.$$

- ▶ So a $(1 - \alpha)100\%$ confidence interval for $\beta_0 + \beta_1 x_{\text{new}}$ is

$$\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \pm \underbrace{t_{n-2, \alpha/2}}_{\hat{t}(1-\alpha/2, n-2)} \underbrace{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{\text{new}} - \bar{x}_n)^2}{S_{xx}}}}_{\text{se}}$$

leverage
of the
point x_{new}



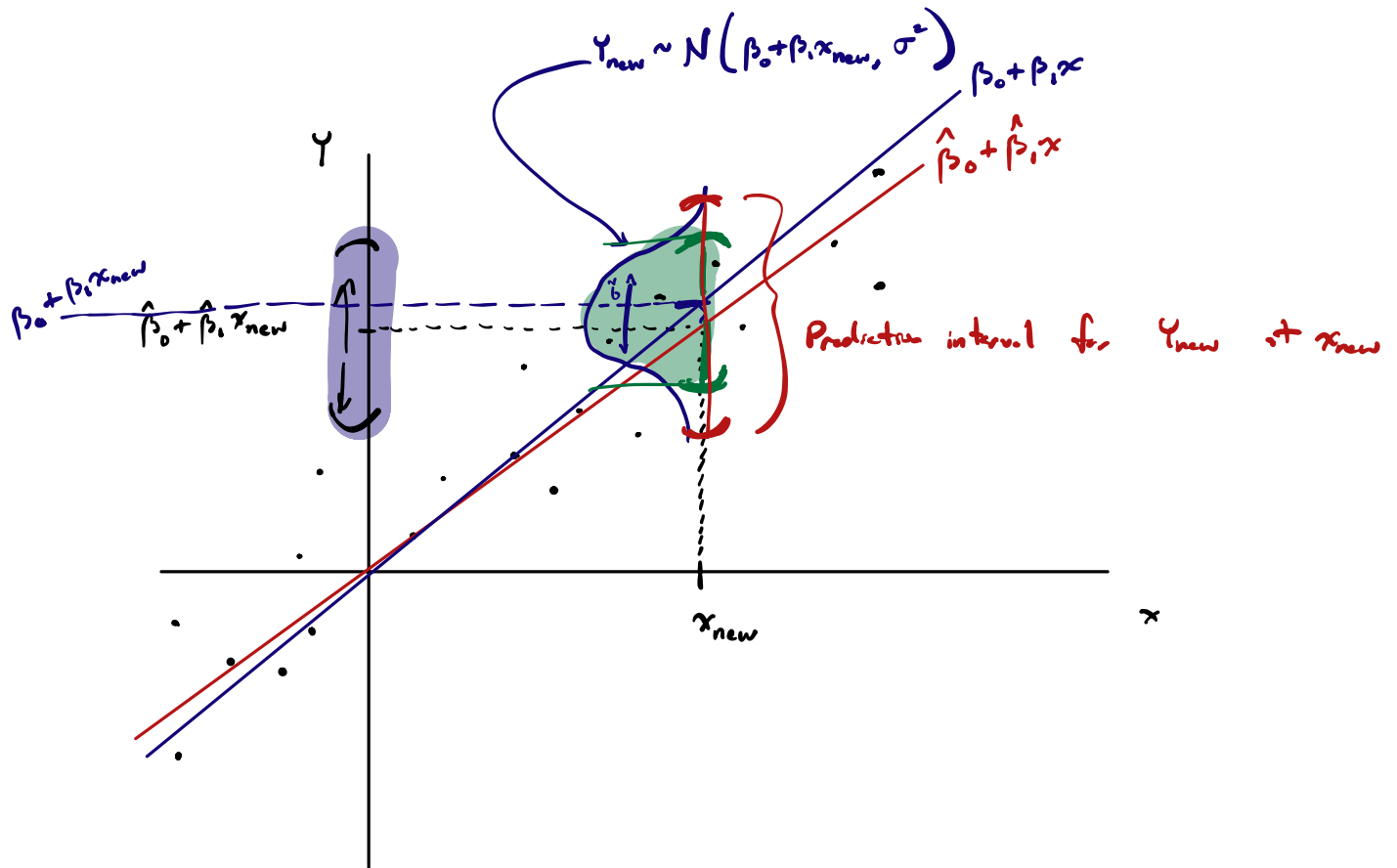
Hemoglobin versus RBC count example (cont)

Give a 95% CI for the mean hemoglobin level of individuals with RBC count 5.5.

```
alpha <- 0.05
xnew <- 5.5
xnew_se <- sqrt(sgsqhat)*sqrt(1/n+(xnew-xbar)^2/Sxx)
ta2 <- qt(1-alpha/2,n-2)
lo <- b0hat + b1hat * xnew - ta2 * xnew_se
up <- b0hat + b1hat * xnew + ta2 * xnew_se
```

We are 95% confident that the mean hemoglobin level of individuals with RBC count 5.5 lies in the interval (14.767, 15.011).

Build a prediction interval for Y_{new} at some x_{new} .



The prediction interval should capture Y_{new} with prob. $1-\alpha$.

Prediction interval for a new value of the response

Assume $\varepsilon_1, \dots, \varepsilon_n \stackrel{\text{ind}}{\sim} \text{Normal}(0, \sigma^2)$. Then:

► The new residual $Y_{\text{new}} - \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}}$ is distributed as

$$\hat{\varepsilon}_{\text{new}} = Y_{\text{new}} - \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \sim \text{Normal} \left(0, \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_{\text{new}} - \bar{x}_n)^2}{S_{xx}} \right] \right).$$

► “Studentizing” the above gives

$$\frac{Y_{\text{new}} - \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}}}{\hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_{\text{new}} - \bar{x}_n)^2}{S_{xx}}}} \sim t_{n-2}.$$

► So a $(1 - \alpha)100\%$ prediction interval for Y_{new} is

$$\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \pm t_{n-2, \alpha/2} \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_{\text{new}} - \bar{x}_n)^2}{S_{xx}}}.$$

↑ only difference from the CI for $\beta_0 + \beta_1 x_{\text{new}}$

Hemoglobin versus RBC count example (cont)

Give a 95% prediction interval for the hemoglobin level when the RBC count is 5.5.

```
alpha <- 0.05
xnew <- 5.5
xnew_pse <- sqrt(sgsqhat)*sqrt(1+1/n+(xnew-xbar)^2/Sxx)
ta2 <- qt(1-alpha/2,n-2)
lo <- b0hat + b1hat * xnew - ta2 * xnew_pse
up <- b0hat + b1hat * xnew + ta2 * xnew_pse
```

We are 95% confident that an *individual* with RBC count 5.5 will have a hemoglobin level in the interval (13.331, 16.447).

Plot confidence and prediction limits over the range of RBC counts.

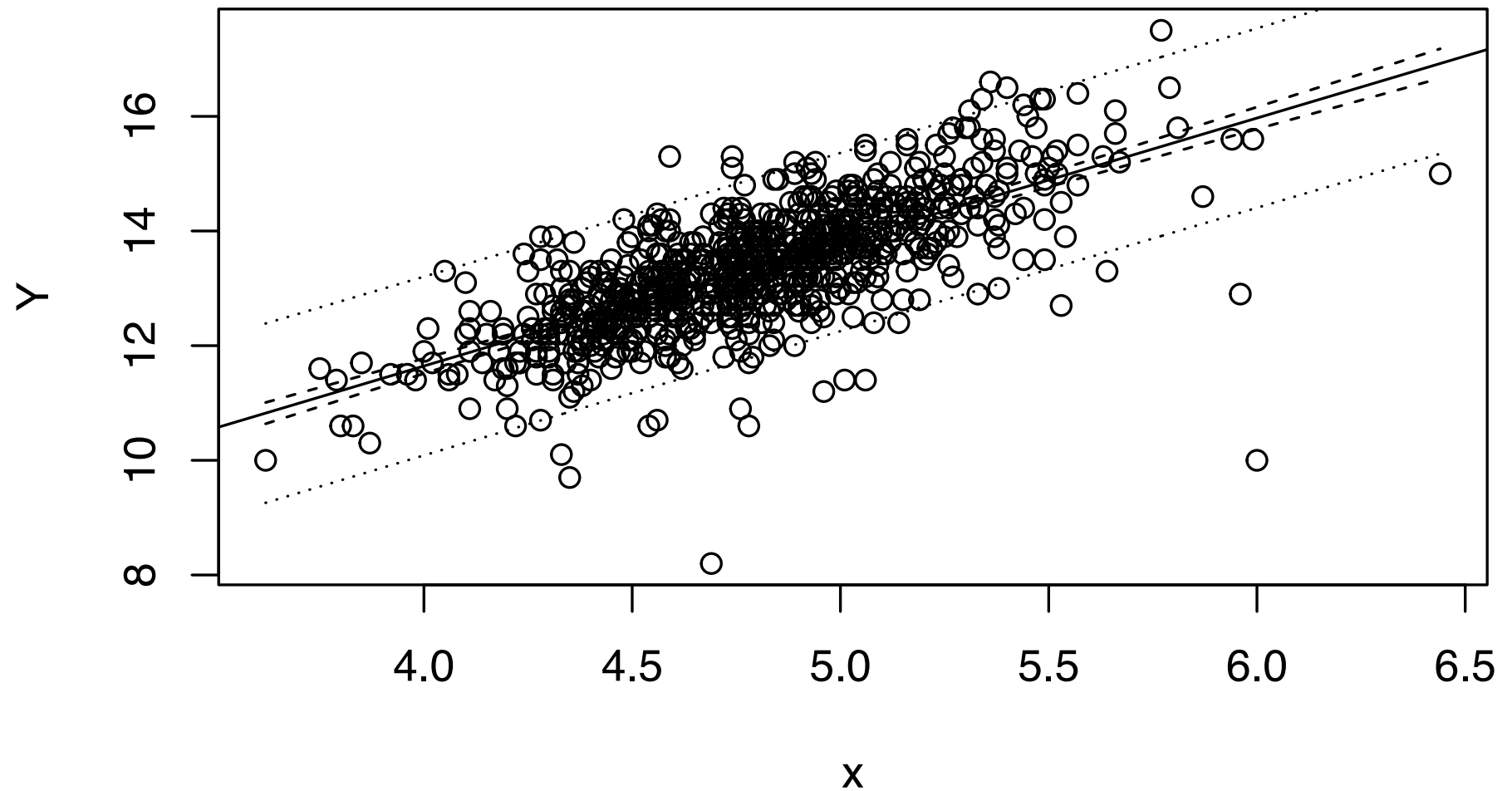
```
alpha <- 0.05
ta2 <- qt(1-alpha/2,n-2)
xseq <- seq(min(x),max(x),length = 500)

xseq_se <- sqrt(sgsqhat)*sqrt(1/n+(xseq-xbar)^2/Sxx)
loci <- b0hat + b1hat * xseq - ta2 * xseq_se
upci <- b0hat + b1hat * xseq + ta2 * xseq_se

xseq_pse <- sqrt(sgsqhat)*sqrt(1+1/n+(xseq-xbar)^2/Sxx)
lopi <- b0hat + b1hat * xseq - ta2 * xseq_pse
uppi <- b0hat + b1hat * xseq + ta2 * xseq_pse
```



```
plot(Y~x)
abline(b0hat,b1hat)
lines(loi ~ xseq, lty=2); lines(upci ~ xseq, lty=2)
lines(lopi ~ xseq, lty=3); lines(upper ~ xseq, lty=3)
```



The predict() function in R

```
predict(lm_out, newdata = data.frame(x = 5.5), int = "conf")
```

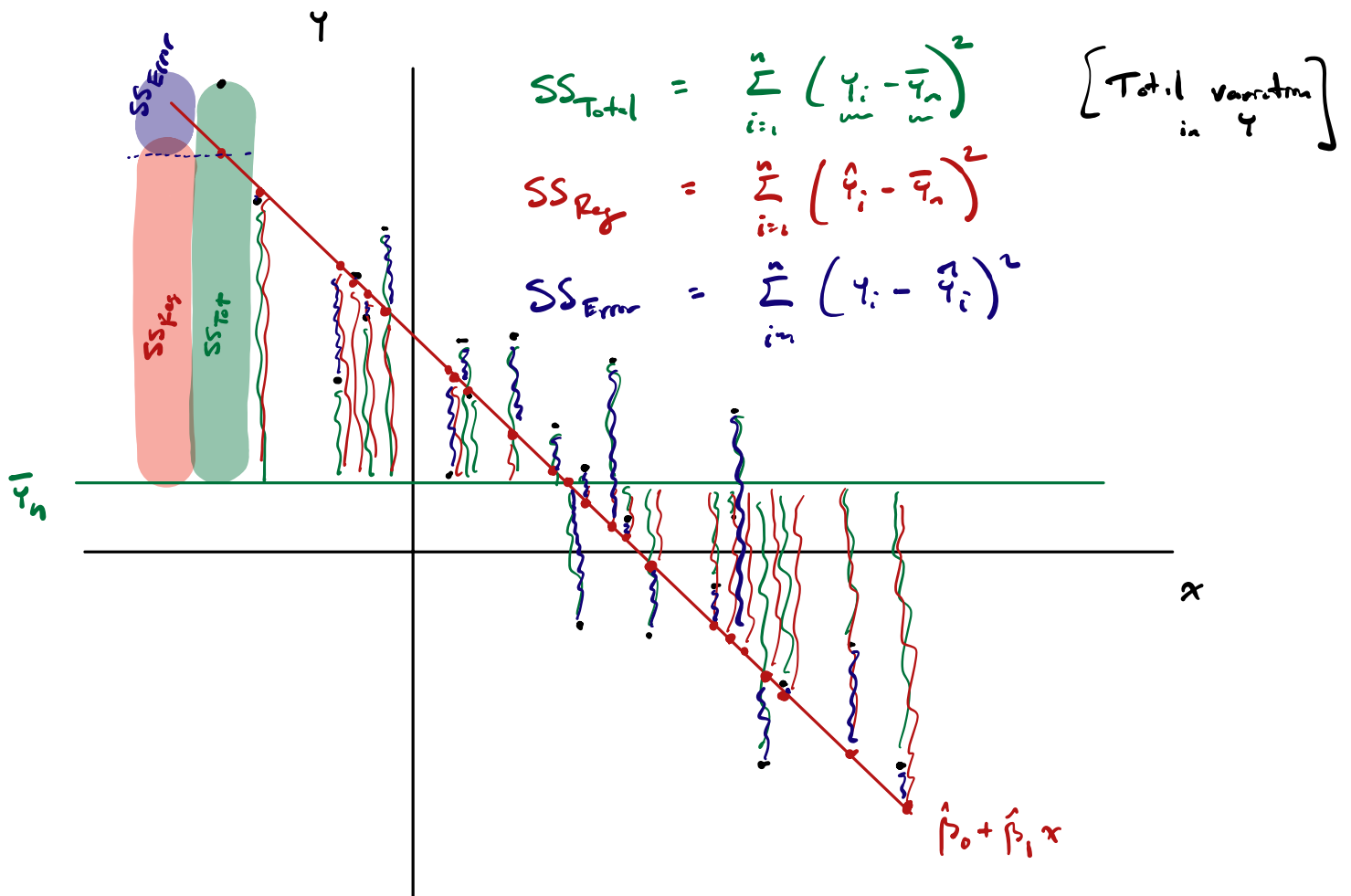
	fit	lwr	upr
1	14.88892	14.76725	15.01059

```
predict(lm_out, newdata = data.frame(x = 5.5), int = "pred")
```

	fit	lwr	upr
1	14.88892	13.33117	16.44667

DECOMPOSE VARIABILITY IN y

Sample variance of y :
$$s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y}_n)^2$$



$$SS_{Total} = SS_{Reg} + SS_{Error}$$

$$R^2 = \frac{SS_{Reg}}{SS_{Total}} \in [0, 1]$$

Sums of squares in simple linear regression

We decompose the variation in $Y_1, \dots, Y_n$ by defining the:

- ▶ Total sum of squares: $SS_{\text{Tot}} = \sum_{i=1}^n (Y_i - \bar{Y}_n)^2$
- ▶ Regression sum of squares: $SS_{\text{Reg}} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y}_n)^2$
- ▶ Error sum of squares: $SS_{\text{Error}} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$

We have $SS_{\text{Tot}} = SS_{\text{Reg}} + SS_{\text{Error}}$.

The coefficient of determination is defined as $R^2 = \frac{SS_{\text{Reg}}}{SS_{\text{Tot}}}$.

Variation in $\hat{Y}$

Total variation
in Y

- ▶ $R^2 \in [0, 1]$
- ▶ Proportion of variation in Y “explained” by the covariate x .
- ▶ In simple linear regression we have $R^2 = r_{xY}^2$.

The mean squares in simple linear regression

The SS, appropriately scaled, follow chi-square distributions:

▶ $\frac{SS_{\text{Tot}}}{\sigma^2} \sim \chi_{n-1}^2(\phi_{\text{Total}})$ ← "phi" = ϕ

▶ $\frac{SS_{\text{Reg}}}{\sigma^2} \sim \chi_1^2(\phi_{\text{Reg}})$

▶ $\frac{SS_{\text{Error}}}{\sigma^2} \sim \chi_{n-2}^2$

The quantities ϕ_{Tot} and ϕ_{Reg} are called noncentrality parameters.

Dividing SS_{Reg} and SS_{Error} by their dfs, we define:

▶ Regression mean square: $MS_{\text{Reg}} = \frac{SS_{\text{Reg}}}{1}$

▶ Error mean square: $MS_{\text{Error}} = \frac{SS_{\text{Error}}}{n - 2}$

The Analysis of Variance (ANOVA) table

Decomposing the variability in Y

We often present the SS, df, and MS values in a table like this:

Source	Df	SS	MS	F value	p-value
Regression	1	SS_{Reg}	MS_{Reg}	F_{stat}	$P(F > F_{\text{stat}})$
Error	$n - 2$	SS_{Error}	MS_{Error}		
Total	$n - 1$	SS_{Tot}			

This is an example of an ANOVA table.

$$MS_{\text{Reg}} = \frac{SS_{\text{Reg}}}{1}$$

$$MS_{\text{Error}} = \frac{SS_{\text{Error}}}{n-2}$$

$$F_{\text{stat}} = \frac{MS_{\text{Reg}}}{MS_{\text{Error}}}$$

Overall F test

SLR: $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$

$$F_{\text{stat}} = \frac{MS_{\text{Reg}}}{MS_{\text{Error}}} = \left(T_{\text{stat}} \right)^2 = \frac{\hat{\beta}_1^2}{\hat{\sigma}^2 / S_{xx}}$$

In addition to the SS, df, and MS value, the ANOVA table presents

► $F_{\text{stat}} = \frac{MS_{\text{Reg}}}{MS_{\text{Error}}}$

► $P(F > F_{\text{stat}})$, where this is computed under $F \sim F_{1, n-2}$

These are the test statistic and p-value of the overall F test.

In simple linear regression this p-value is the same as the one for testing $H_0: \beta_1 = 0$ versus $H_1: \beta_1 \neq 0$ with the t test; moreover $F_{\text{stat}} = T_{\text{stat}}^2$.

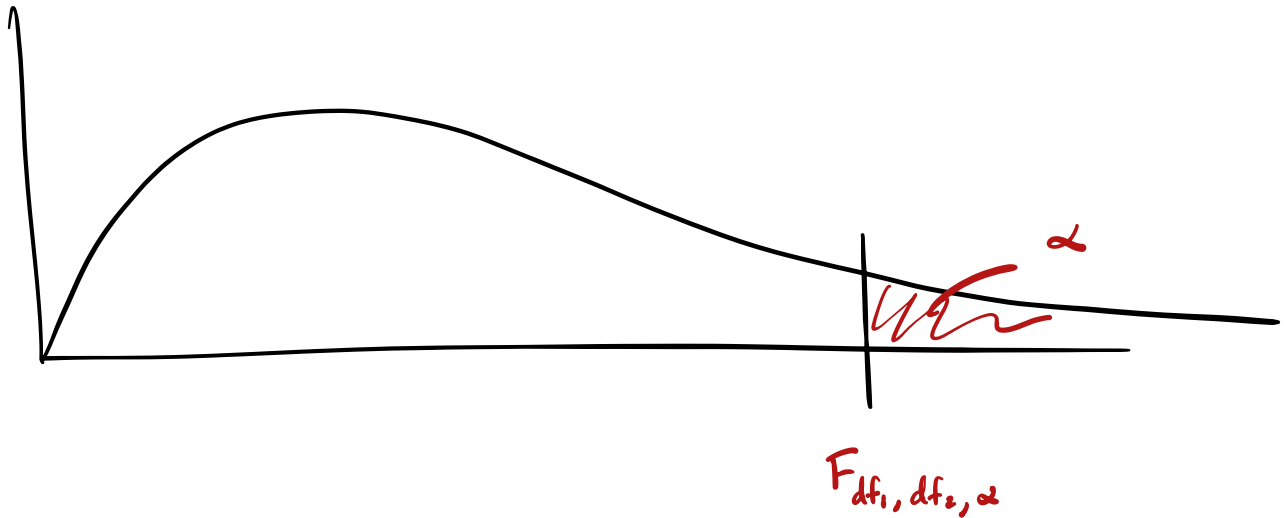
For what it's worth, one can show $F_{\text{stat}} = \frac{(n-2)r_{xY}^2}{1 - r_{xY}^2}$ in SLR.

We will discuss the overall F test in greater detail later.

↑ Misc.

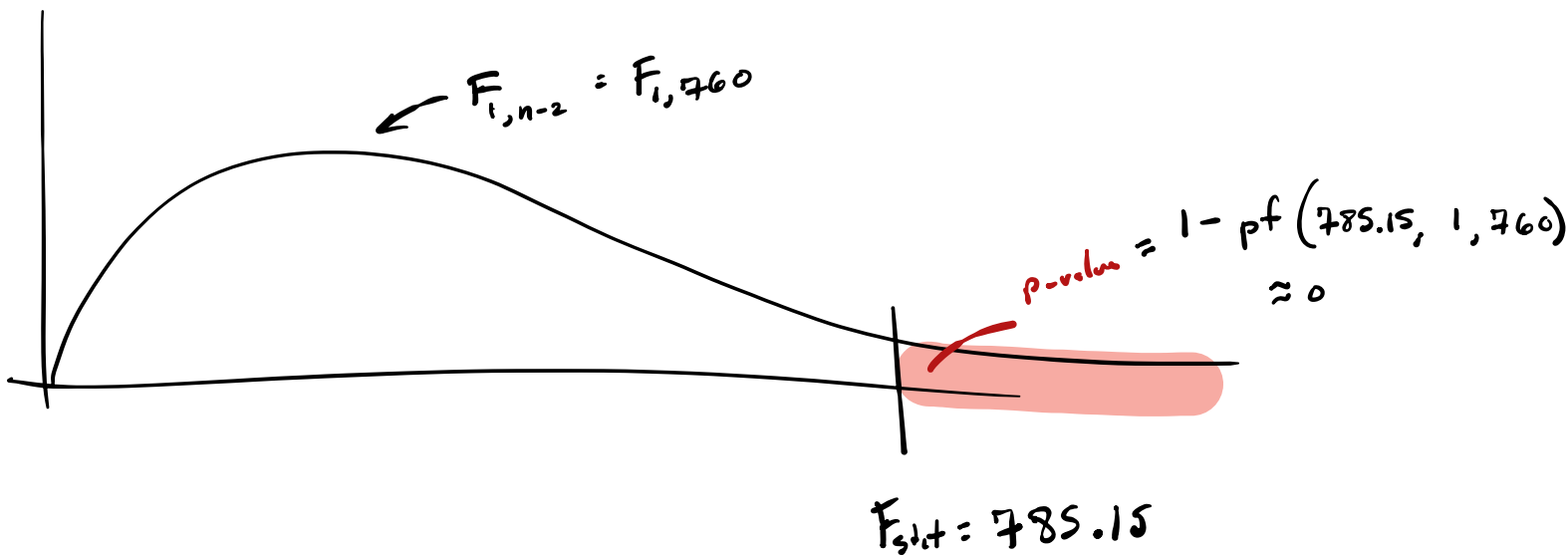
$$F_{df_1, df_2} = \frac{\chi^2_{df_1} / df_1}{\chi^2_{df_2} / df_2} \quad \chi^2_{df_1}, \chi^2_{df_2} \text{ independent}$$

$\uparrow$ "numerator" $\uparrow$ "denominator"



For Hem / RBL

$$H_0: \beta_1 = 0 \quad \text{vs} \quad H_1: \beta_1 \neq 0$$



Building the ANOVA table

```
SST <- sum((Y - Ybar)^2)
SSR <- sum((Yhat - Ybar)^2)
SSE <- sum((Y - Yhat)^2)
MSR <- SSR / 1
MSE <- SSE / (n-2)
Fstat <- MSR / MSE # same as (n-2)*rxY^2/(1 - rxY^2)
pval <- 1 - pf(Fstat,1,n-2)
```

Source	Df	SS	MS	F value	p-value
x	1	491.37	491.37	785.15	0
Error	760	475.63	0.63		
Total	761	967			

$n-2$

$n-1$

$$\frac{491.37}{0.63} = 785.15$$

$$475.63 / 760$$

The `lm()`, `summary()`, and `anova()` functions in R

```
lm_out <- lm(Y~x)
lm_out
```

Call:

```
lm(formula = Y ~ x)
```

Coefficients:

(Intercept)	x
2.994	2.163

```
summary(lm_out)
```

Call:

```
lm(formula = Y ~ x)
```

$$28.021 = \frac{2.163}{0.0772}$$

Residuals:

Min	1Q	Median	3Q	Max
-5.9702	-0.4232	0.0074	0.4645	2.3791

$$\hat{\sigma} / \sqrt{S_{xx}}$$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.99447	0.37065	8.079	2.56e-15 ***
x	2.16263	0.07718	28.021	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7911 on 760 degrees of freedom

Multiple R-squared: 0.5081, Adjusted R-squared: 0.5075

F-statistic: 785.1 on 1 and 760 DF, p-value: < 2.2e-16

$$F_{stat} = \frac{MS_{Reg}}{MS_{Error}} \quad df_1 \quad R^2 = \frac{SS_{Reg}}{SS_{Total}} \quad df_2$$

```
anova(lm_out)
```

Analysis of Variance Table

Response: Y

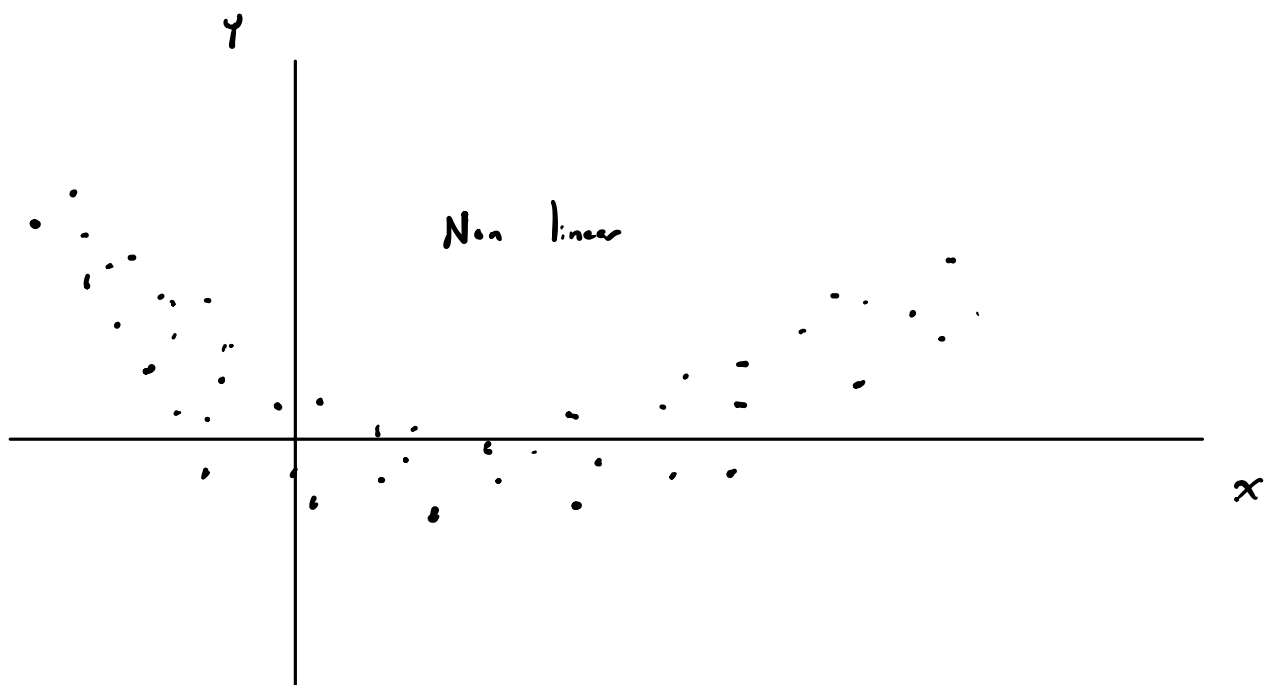
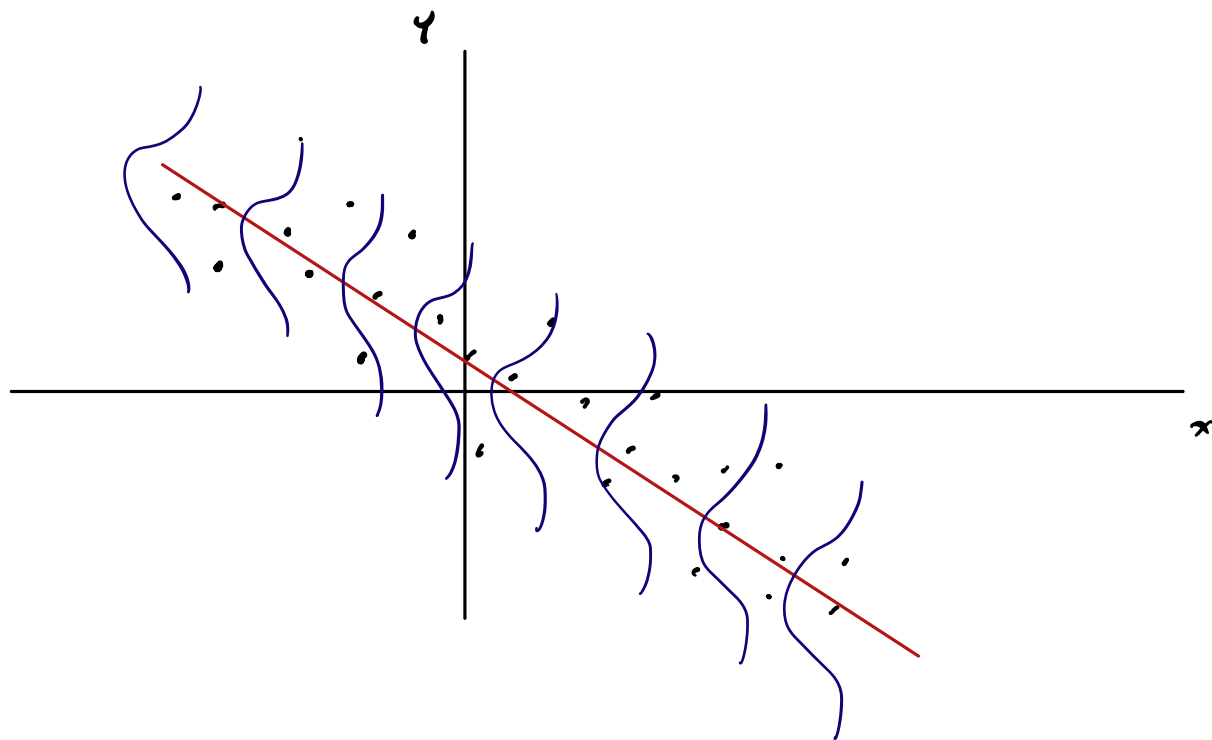
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
x	1	491.37	491.37	785.15	< 2.2e-16 ***
Residuals	760	475.63	0.63		

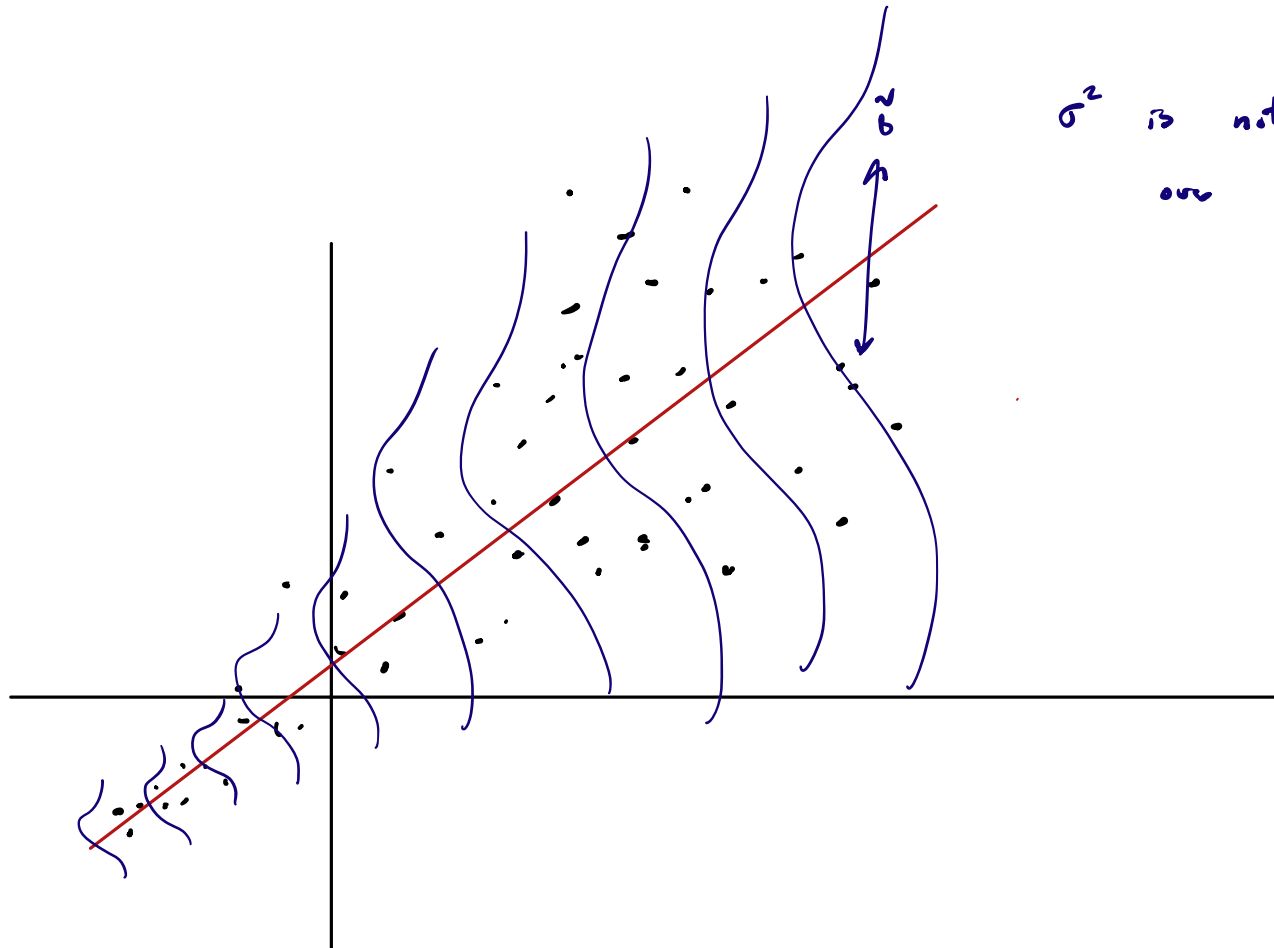
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Observe $(x_1, y_1), \dots, (x_n, y_n)$ such that

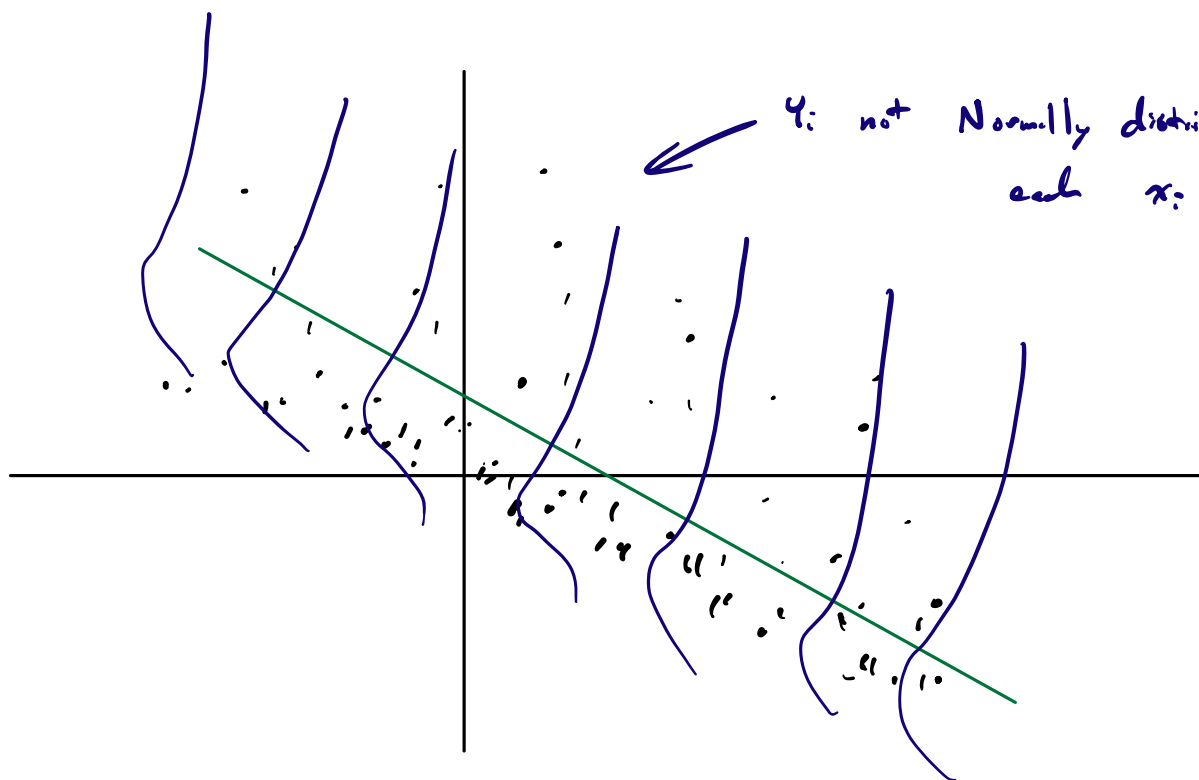
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad i = 1, \dots, n$$

where $\varepsilon_1, \dots, \varepsilon_n \overset{\text{independent}}{\sim} N(0, \sigma^2)$.





σ^2 is not constant
over the x values



y_i not Normally distributed at
each x_i

Checking model assumptions

Validity of the foregoing analyses depends on these assumptions:

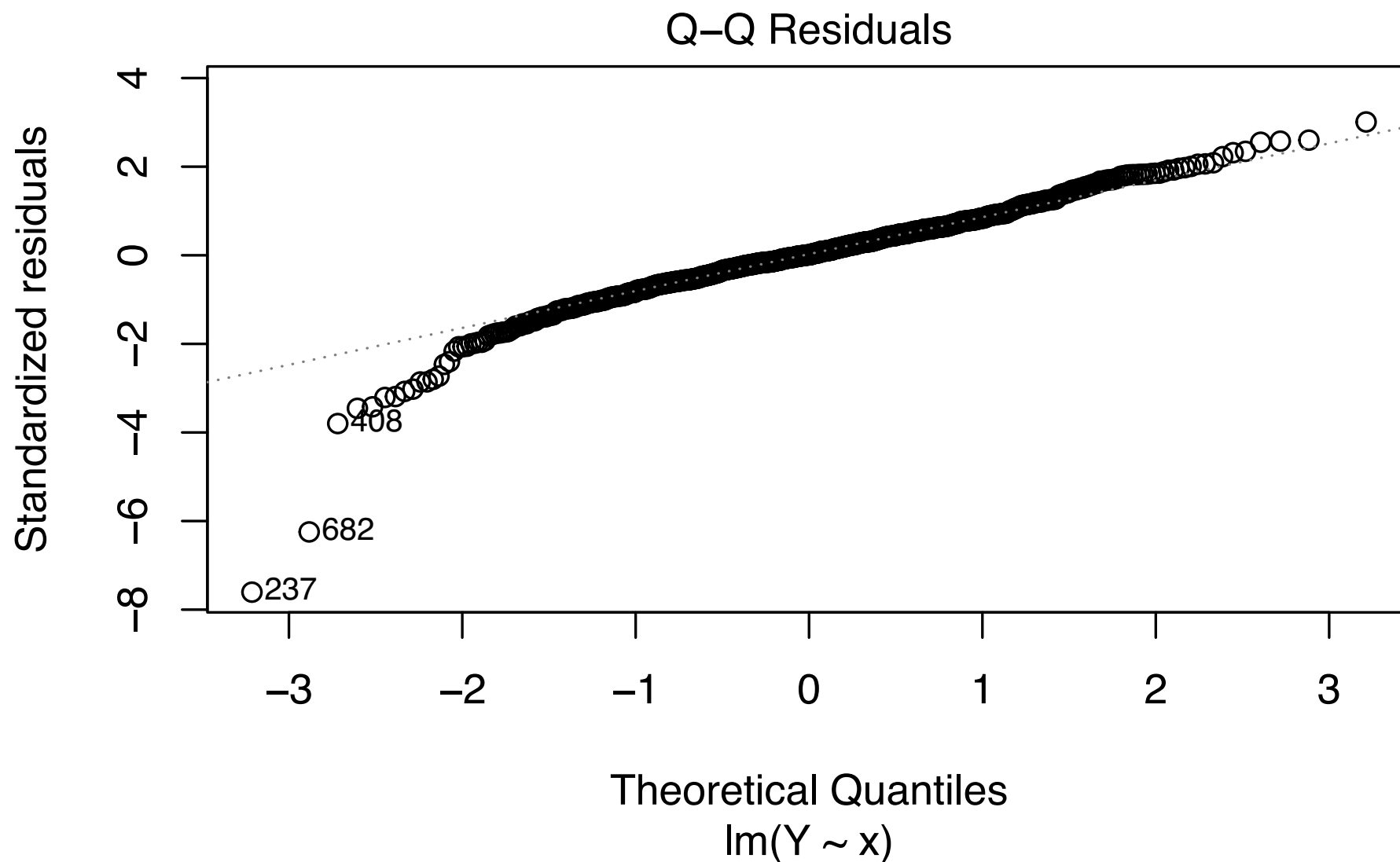
1. The responses are normally distributed around the regression line (Check QQ plot of residuals). *If n is large this doesn't matter.*
2. The response has the same variance for all values of the covariate (Check residuals vs fitted values plot). *plot of $(\hat{\varepsilon}_i, \hat{y}_i)$, $i = 1, \dots, n$*
3. The covariate and the response are linearly related (Check residuals vs fitted values plot).
4. The response values are independent of each other (No way to check; must trust experimental design).

Residuals

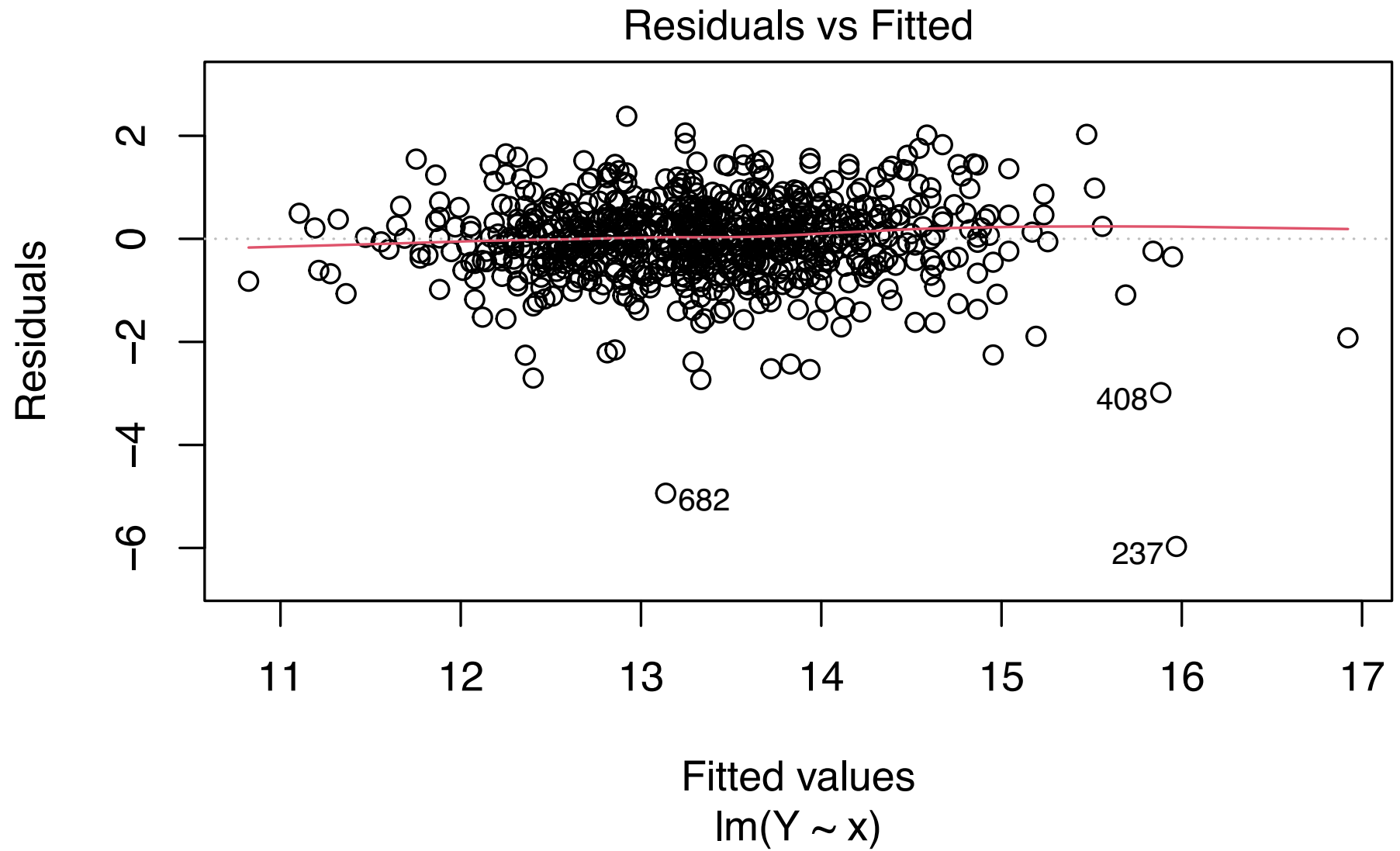
$$\hat{\varepsilon}_i = \underset{\substack{\uparrow \\ \text{obs}}}{y_i} - \underset{\substack{\nwarrow \\ \text{fitted}}}{\hat{y}_i}$$

Generating diagnostic plots from `lm()` with `plot()`

```
plot(lm_out, which = 2)
```




```
plot(lm_out, which = 1)
```



The abalone...



Photo on the left by Sharktopus - Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=14082271>

Abalone data example

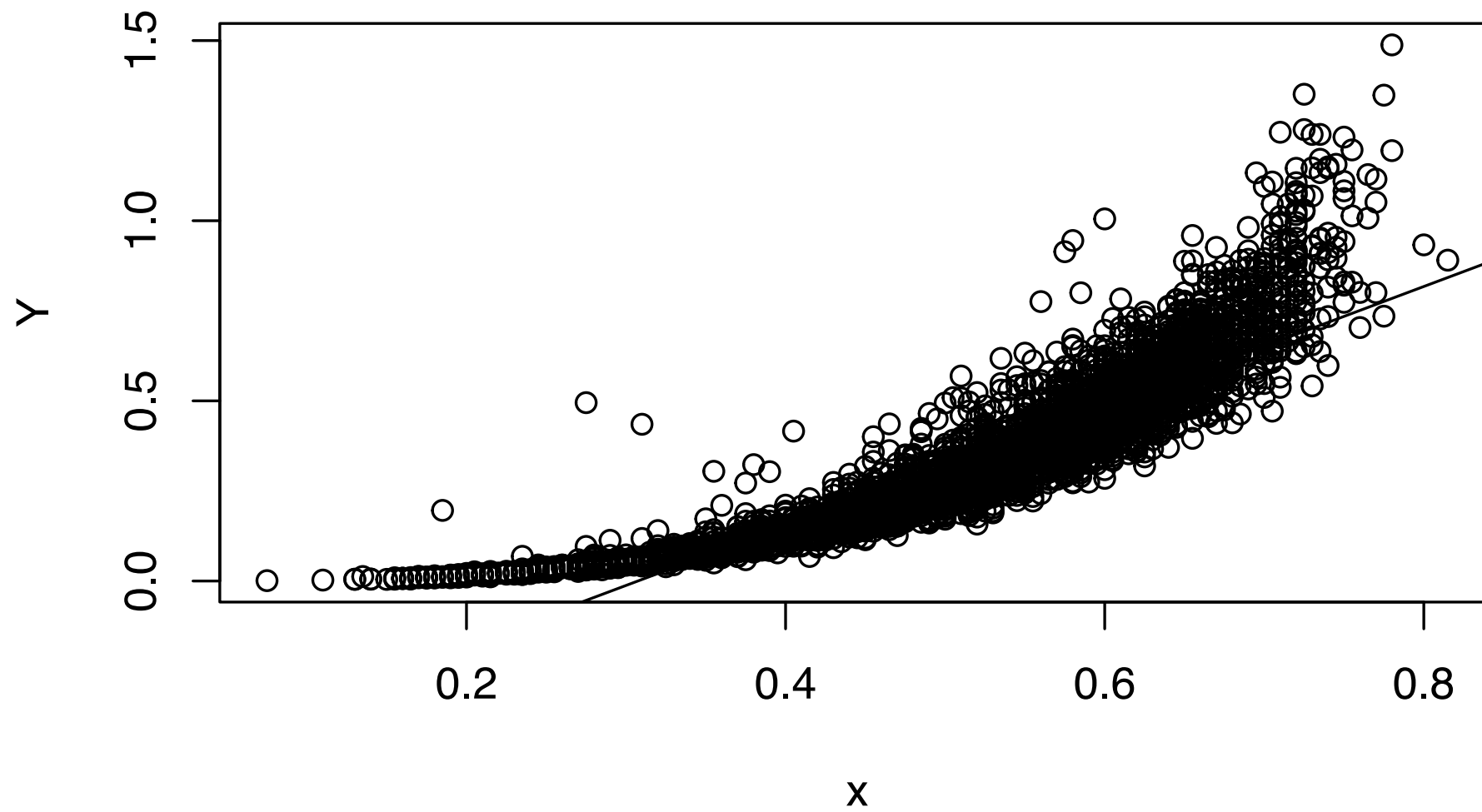
Predict shucked weight of an abalone by its length.

```
csv <- url("https://people.stat.sc.edu/gregorkb/data/abalone.csv")
abalone <- read.csv(csv,col.names = c("Sex",
                                      "Length",
                                      "Diameter",
                                      "Height",
                                      "Whole_Wt",
                                      "Shucked_Wt",
                                      "Viscera_Wt",
                                      "Shell_Wt",
                                      "Rings"))

Y <- abalone$Shucked_Wt
x <- abalone$Length
n <- length(Y)
```

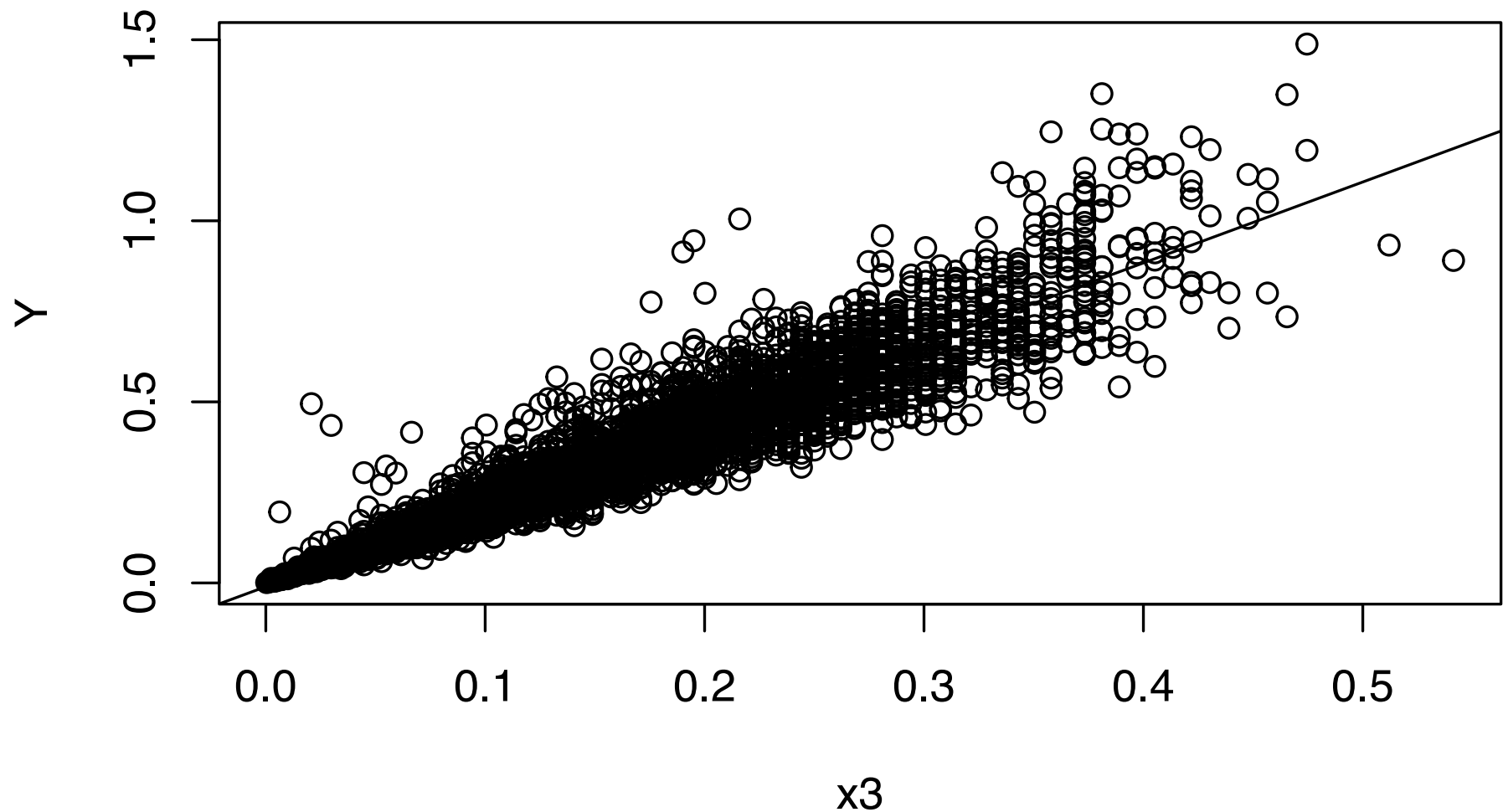
There are $n = 4176$ records. Data come from Nash and Ford (1995).

```
lm1 <- lm(Y~x)
plot(Y~x)
abline(lm1)
```

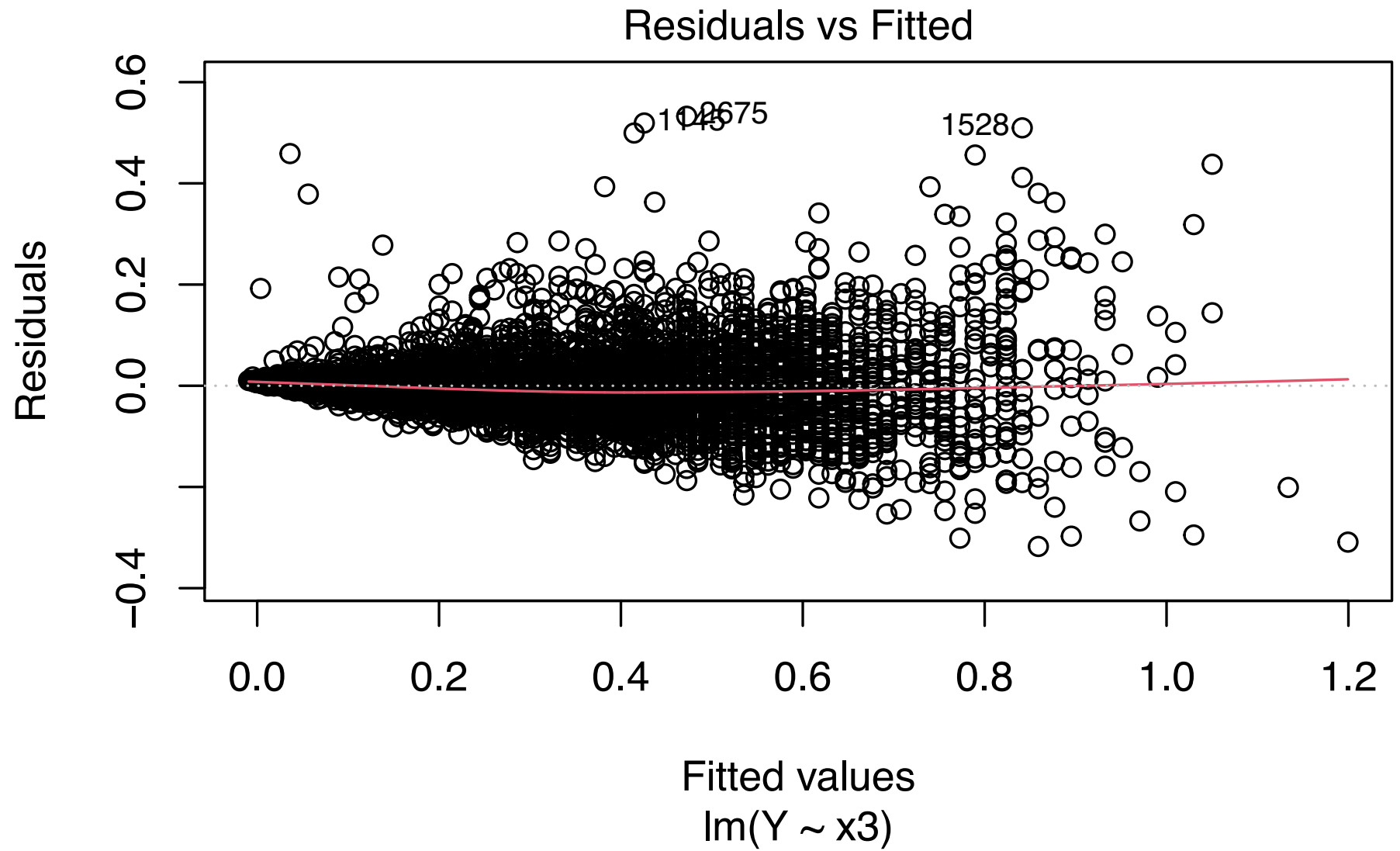


Try transforming x:

```
x3 <- x**3  
lm2 <- lm(Y ~ x3)  
plot(Y ~ x3); abline(lm2)
```

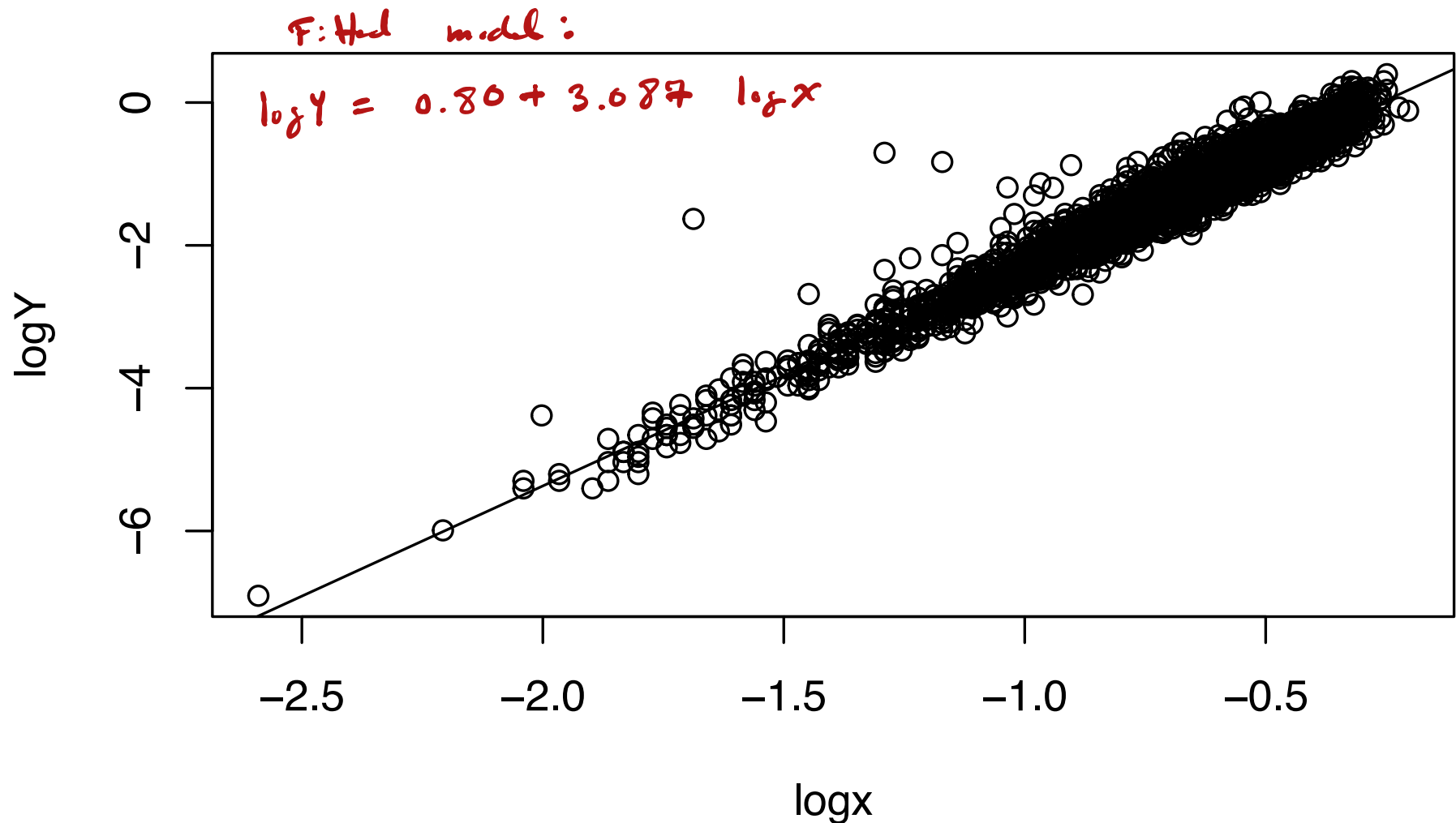


```
plot(lm2, which = 1)
```

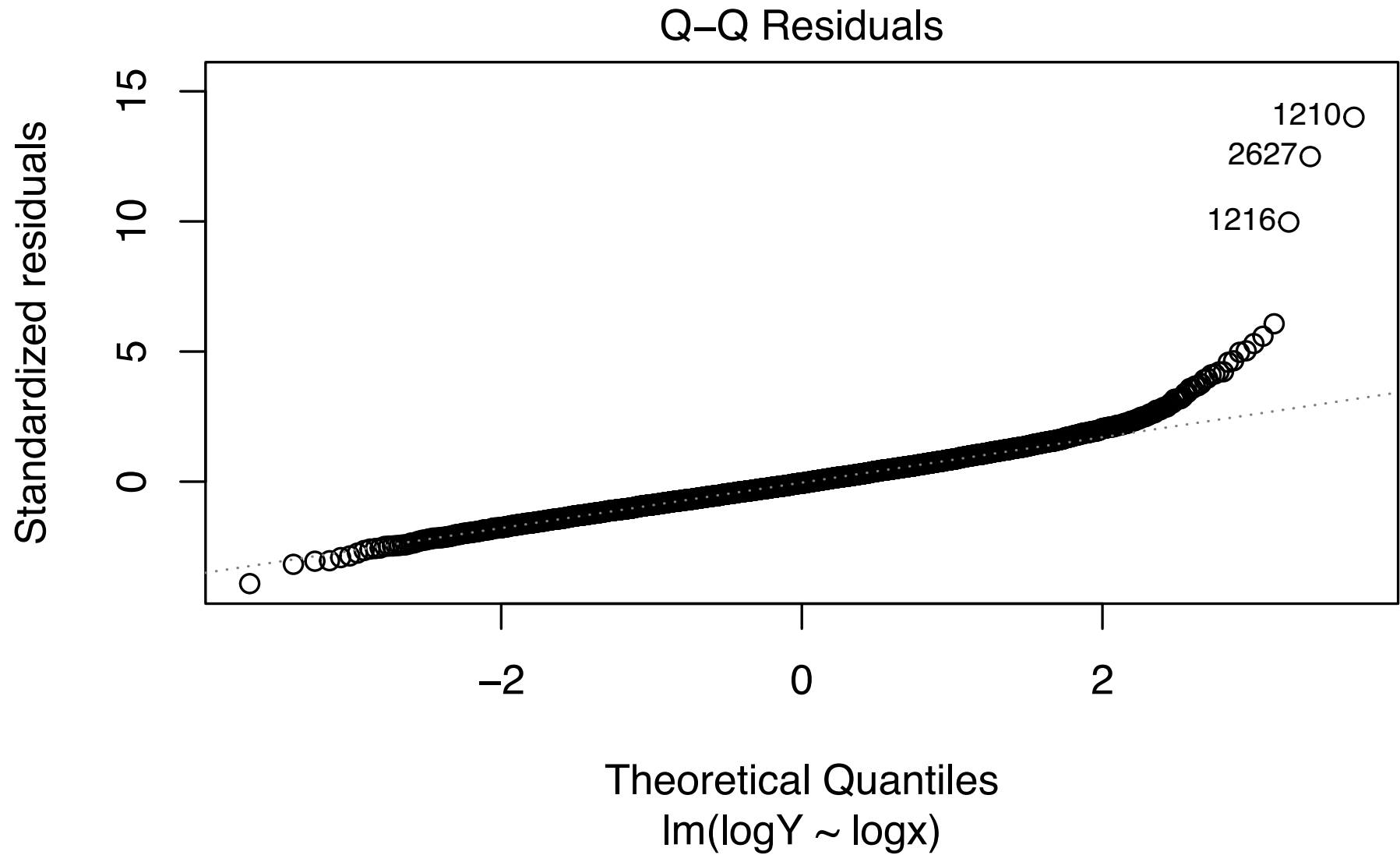


Could transform both Y and x:

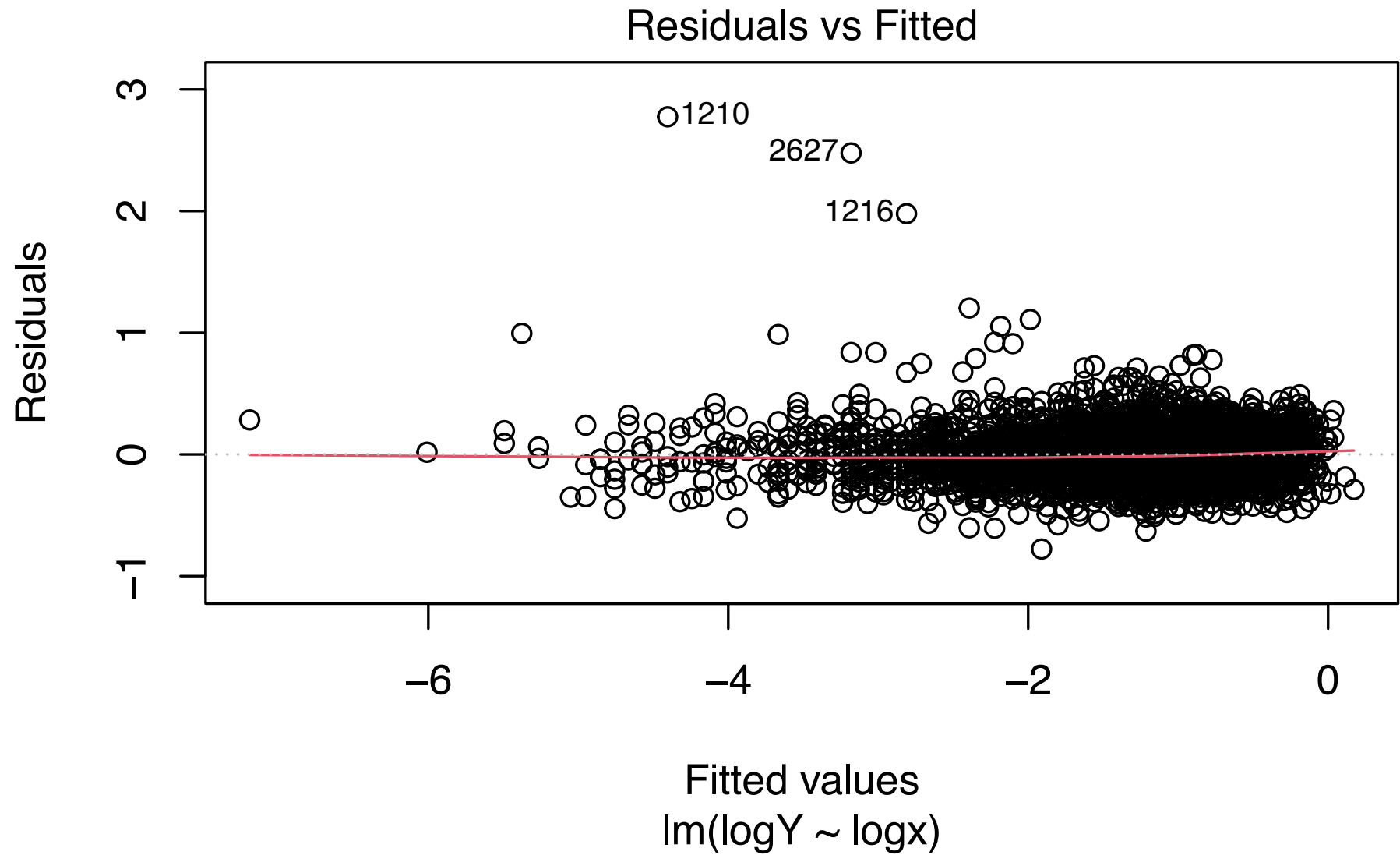
```
logY <- log(Y); logx <- log(x)  
lm3 <- lm(logY ~ logx)  
plot(logY~logx); abline(lm3)
```



```
plot(lm3, which = 2)
```




```
plot(lm3, which = 1)
```



Transforming variables to obtain a linear relationship

Take care how to interpret β_1 after transforming the data.

Example: Log transforming x and Y gives β_1 a %-change interpretation:

$$\log y = \beta_0 + \beta_1 \log x \iff \frac{d \log y}{dx} = \beta_1 \frac{1}{x} \iff \frac{dy}{y} = \beta_1 \frac{dx}{x}$$

Abalone data example (cont)

We must back-transform prediction intervals if we have transformed Y .

```
xnew <- 0.5
newdata <- data.frame( logx = log(xnew))
pi_logY <- predict(lm3,newdata = newdata, int = "pred")
pi_logY
```

	fit	lwr	upr
1	-1.335636	-1.724831	-0.946441

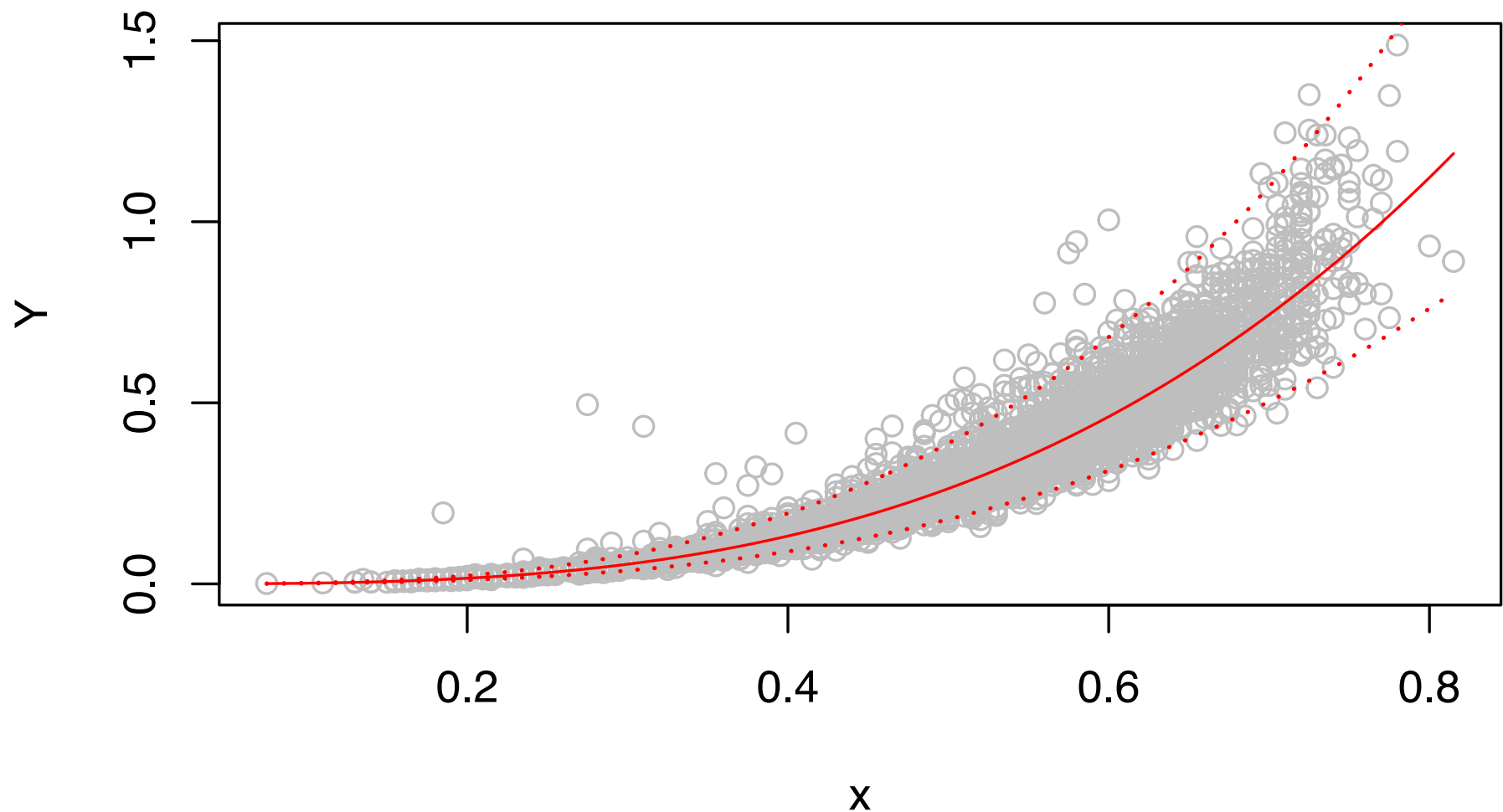
```
pi <- exp(pi_logY)
pi
```

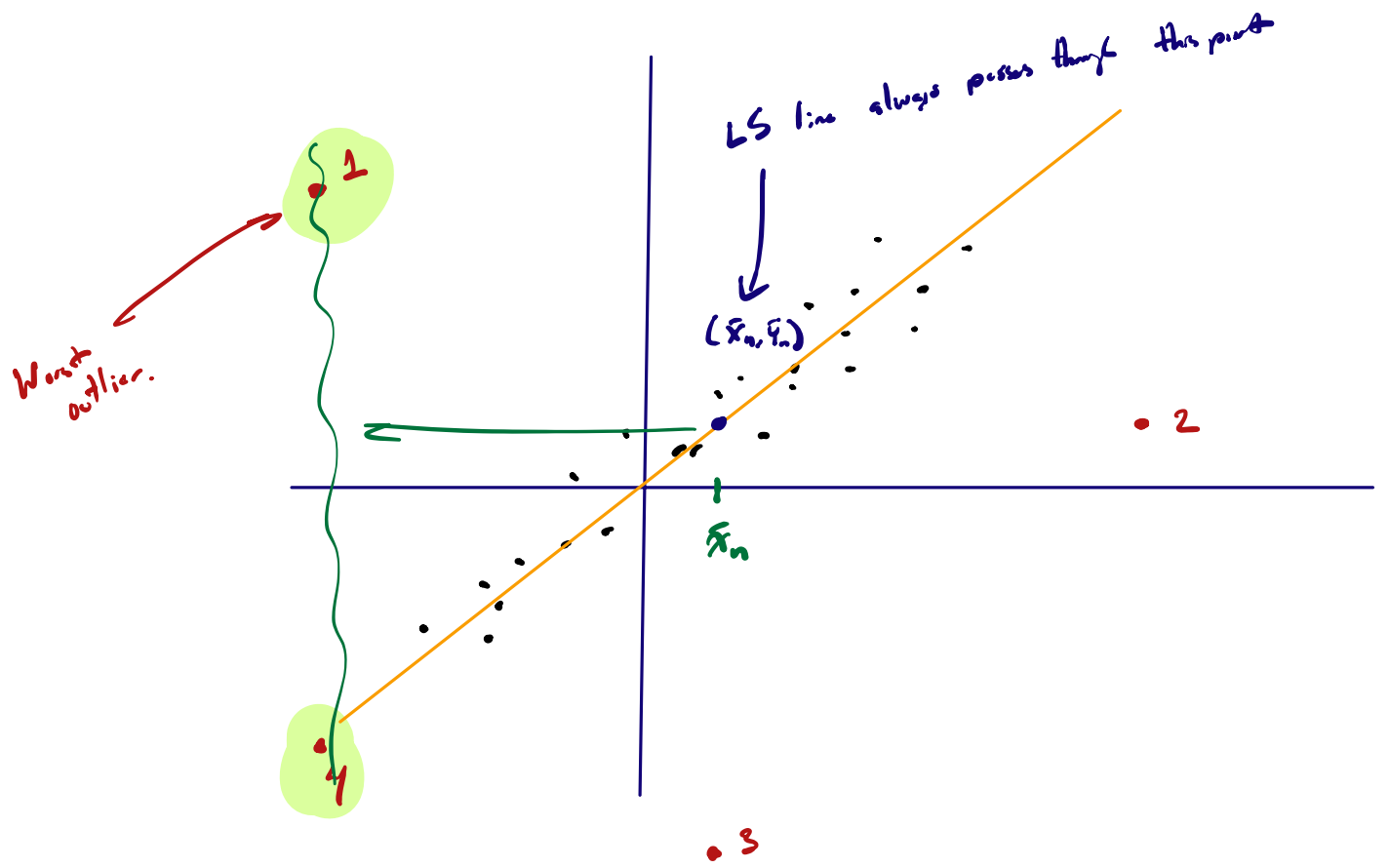
	fit	lwr	upr
1	0.2629909	0.1782032	0.3881199

```

plot(Y~x,col="gray"); logx <- seq(min(logx),max(logx),length=500)
newdata <- data.frame(logx = logx)
logy_hat <- predict(lm3,newdata = newdata,int = "pred")
lines(exp(logy_hat[,1]) ~ exp(logx), col = "red")
lines(exp(logy_hat[,2]) ~ exp(logx), col = "red", lty = 3,lwd=1.5)
lines(exp(logy_hat[,3]) ~ exp(logx), col = "red", lty = 3,lwd=1.5)

```





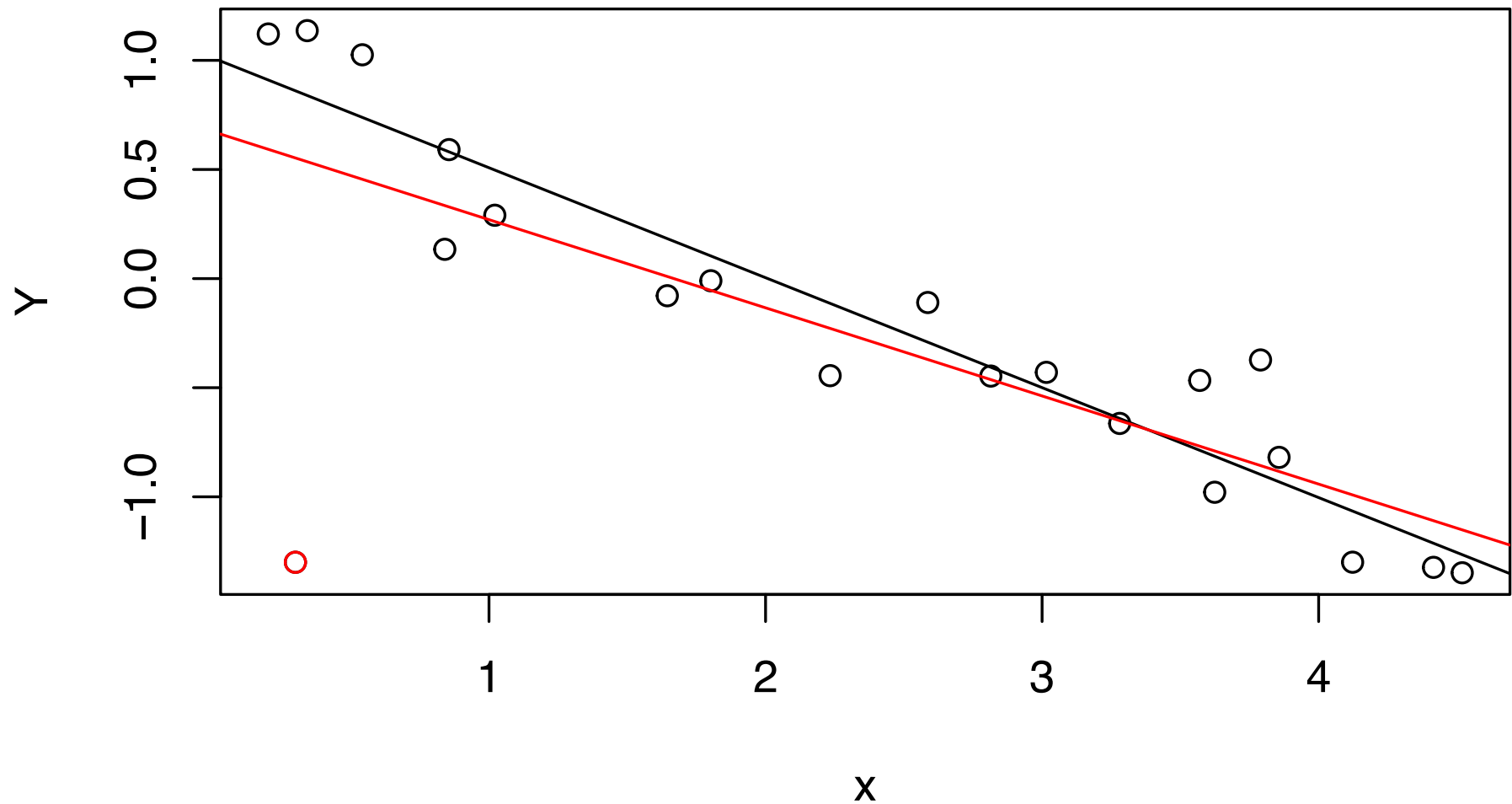
Outliers in simple linear regression

Outlying data points can have a large influence on the estimated regression function.

Let's generate some data and then add an outlier:

```
n <- 20
b0 <- 1
b1 <- -1/2
sg <- .2
x0 <- runif(n,0,5)
e <- rnorm(n,0,sg)
Y0 <- b0 + b1 * x0 + e
x <- c(x0,.3)
Y <- c(Y0,-1.3)
```

```
plot(Y~x);points(Y[n+1]~x[n+1], col = "red")  
abline(lm(Y0~x0))  
abline(lm(Y~x), col = "red")
```



The red data point appears to exert an undue influence over the fit.

Leverage and Cook's distance

How far is x_i from the mean $\bar{x}$ of $x_1, \dots, x_n$

The leverage of a point (x_i, Y_i) among $(x_1, Y_1), \dots, (x_n, Y_n)$ is

$$\text{lev}_i = \frac{1}{n} + \frac{(x_i - \bar{x}_n)^2}{S_{xx}}$$

Leverage only shows outlying-ness in the x direction.

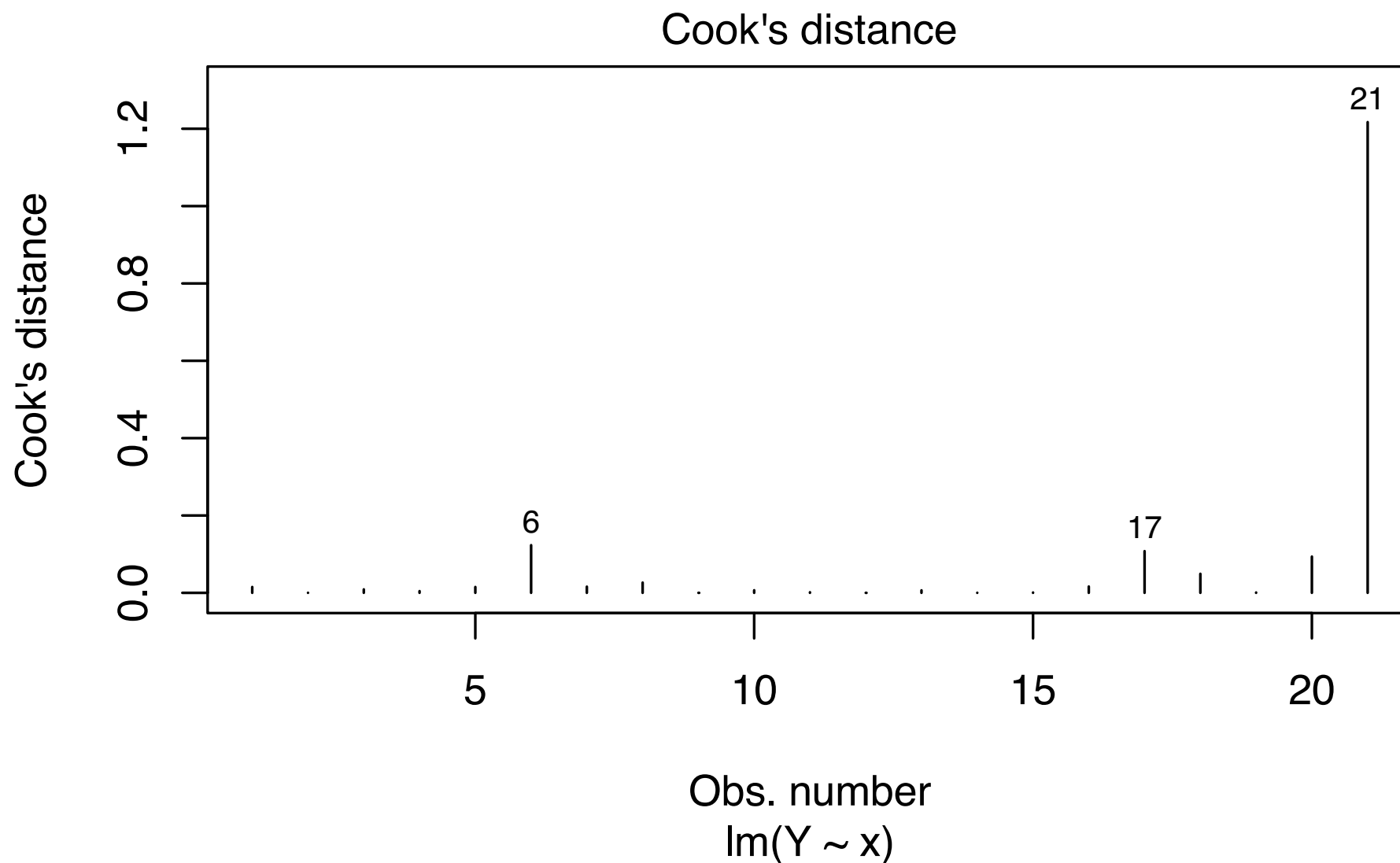
Cook's Distance measures how much each data point changes the fit:

$$D_i = \frac{1}{2\hat{\sigma}^2} \sum_{j=1}^n (\hat{Y}_j - \hat{Y}_{j(i)})^2 = \frac{\hat{e}_i^2}{2\hat{\sigma}^2} \frac{\text{lev}_i}{(1 - \text{lev}_i)^2} \quad \text{for } i = 1, \dots, n,$$

where $\hat{Y}_{j(i)}$ is the j th fitted value from the model fitted without obs i .

Make a plot of the Cook's distances

```
plot(lm(Y~x),which = 4)
```

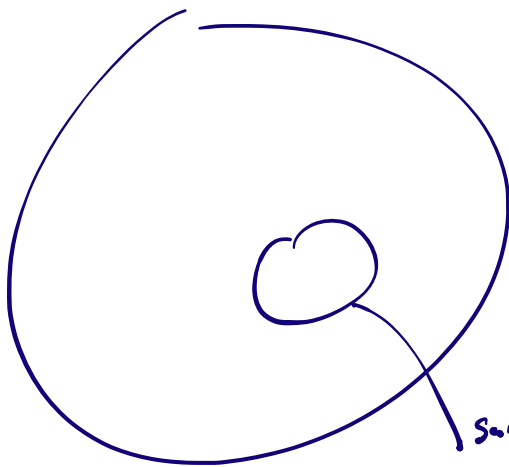


Code to compute Cook's distances

```
n <- length(Y)
xbar <- mean(x)
Ybar <- mean(Y)
rxY <- cor(x,Y) # Pearson's correlation coefficient
b1hat <- rxY * sd(Y)/sd(x)
b0hat <- Ybar - b1hat * xbar
Sxx <- sum((x - xbar)^2)
lev <- 1/n + (x - xbar)^2/Sxx
Yhat <- b0hat + b1hat * x
ehat <- Y - Yhat
sgsqhat <- sum(ehat^2)/(n-2)

cooksD <- ehat^2 / (2*sgsqhat) * lev / (1 - lev)^2
```

pop 1: μ_1, σ_1^2



Sample

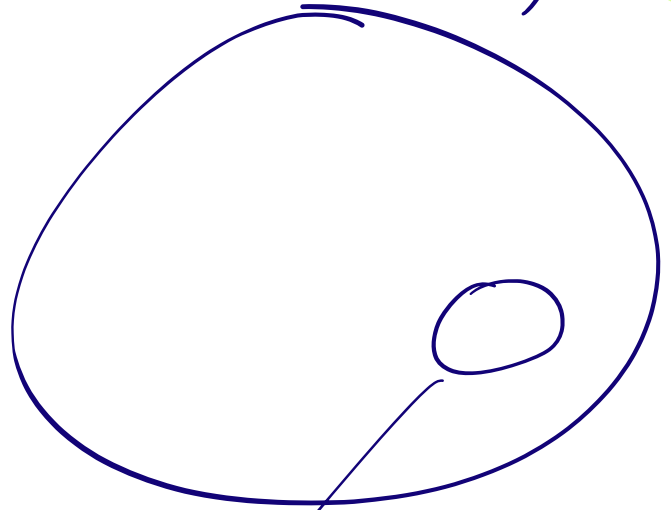
$\bar{x}_1, s_1^2$

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

Assume $\sigma_1^2 = \sigma_2^2$.

pop 2: μ_2, σ_2^2



$\bar{x}_2, s_2^2$

Two-sample t-test by simple linear regression

Let $Y_{ij} \stackrel{\text{ind}}{\sim} \text{Normal}(\mu_i, \sigma^2)$, $j = 1, \dots, n_i$, $i = 1, 2$ and consider

$$H_0: \mu_2 - \mu_1 = 0 \text{ versus } H_1: \mu_2 - \mu_1 \neq 0.$$

The (equal-variances) two-sample t-test uses the test statistic

$$T_{\text{stat}} = \frac{\bar{Y}_2 - \bar{Y}_1}{S_{\text{pooled}} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where $\bar{y}_i = n_i^{-1} \sum_{j=1}^{n_i} Y_{ij}$, $i = 1, 2$ and

$$S_{\text{pooled}}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}, \quad S_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2.$$

We reject H_0 at significance level α if $|T_{\text{stat}}| > t_{n-2, \alpha/2}$.

Appendicitis example

Look again at the data from Marcinkevičs et al. (2023).

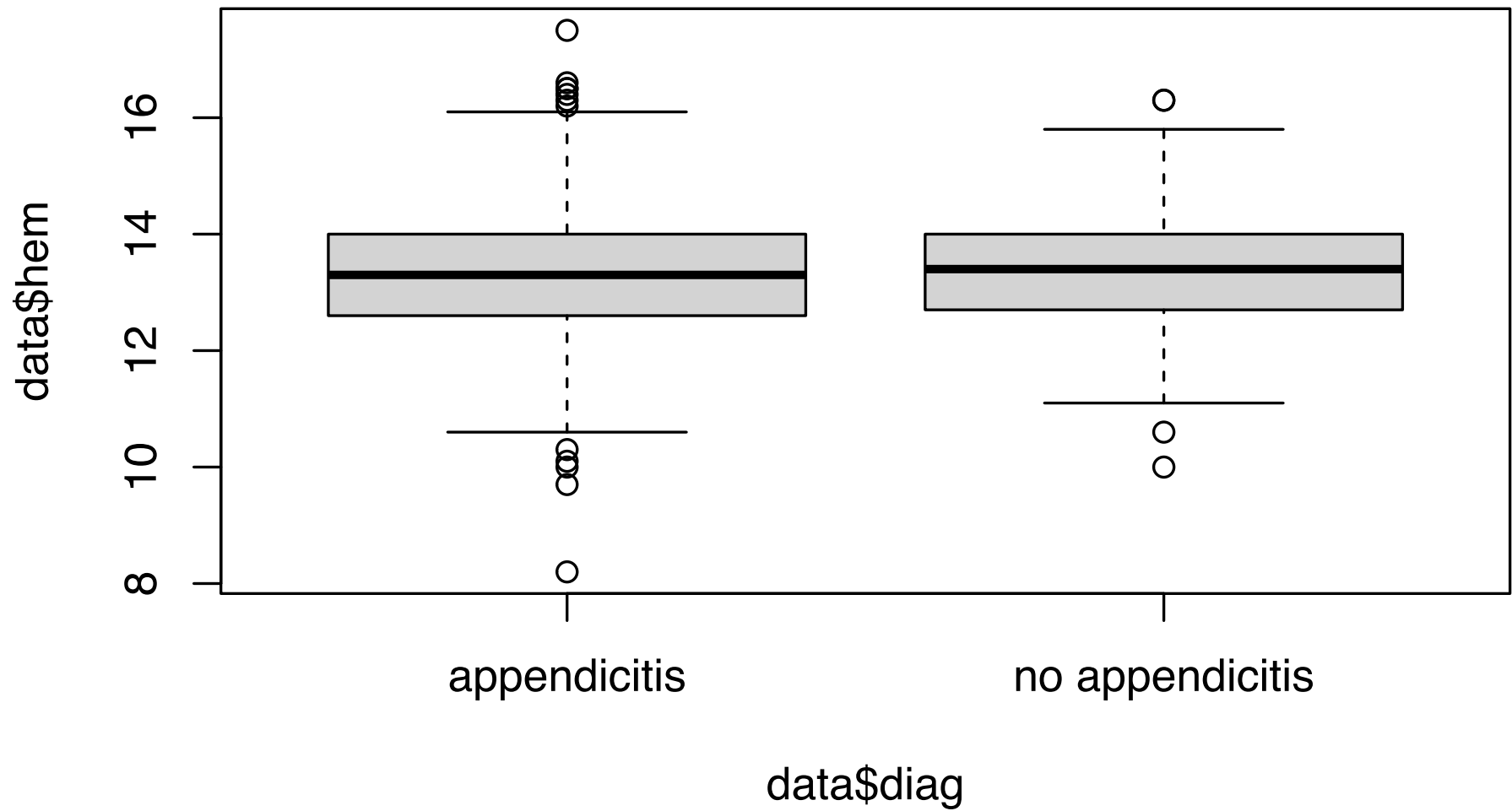
```
link <- url("https://people.stat.sc.edu/gregorkb/data/hrbc.csv")
data <- read.csv(link)
head(data)
```

	hem	rbc	sex	age	diag
1	14.8	5.27	female	12.68	appendicitis
2	15.7	5.26	male	14.10	no appendicitis
3	11.4	3.98	female	14.14	no appendicitis
4	13.6	4.64	female	16.37	no appendicitis
5	12.6	4.44	female	11.08	appendicitis
6	12.5	4.96	male	11.05	no appendicitis

Is the mean hemaglobin level the same in children with and without appendicitis (ignoring rbc, age, and sex)?

Appendicitis example (cont)

```
boxplot(data$hem ~ data$diag)
```



Appendicitis example (cont)

```
t.test(data$hem ~ data$diag, var.equal = TRUE)
```

Two Sample t-test

data: data\$hem by data\$diag

t = -0.49212, df = 760, p-value = 0.6228

alternative hypothesis: true difference in means between group appendicitis and

95 percent confidence interval:

-0.2038964 0.1221585

sample estimates:

mean in group appendicitis	mean in group no appendicitis
----------------------------	-------------------------------

13.33229	13.37316
----------	----------

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad i = 1, \dots, n$$

$$x_i = \begin{cases} 1 & \text{if "appendicitis"} \\ 0 & \text{if "no appendicitis"} \end{cases}$$

$$E Y_i = E(\beta_0 + \beta_1 x_i + \varepsilon_i) = \beta_0 + \beta_1 x_i$$

$$E Y_i = \begin{cases} \beta_0 + \beta_1 & \text{if "appendicitis"} \\ \beta_0 & \text{if "no appendicitis"} \end{cases}$$

Then we can write

$$\mu_1 = \beta_0 + \beta_1 \quad (\text{appendicitis})$$

$$\mu_2 = \beta_0 \quad (\text{no appendicitis})$$

So then $\beta_1 = \mu_1 - \mu_2$.

I can test $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$.

Appendicitis example (cont)

Let the Y_i be the hemaglobin values and define an indicator variable as

$$x_i = \begin{cases} 0 & \text{if no appendicitis} \\ 1 & \text{if appendicitis} \end{cases} \quad \text{for } i = 1, \dots, n.$$

Then in the SLR model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ we have

- ▶ $\beta_0 = \mu_{\text{no app}}$
- ▶ $\beta_0 + \beta_1 = \mu_{\text{app}}$
- ▶ $\beta_1 = \mu_{\text{app}} - \mu_{\text{no app}}$

The t test in the simple linear regression setup of

$$H_0: \beta_1 = 0 \text{ versus } H_1: \beta_1 \neq 0$$

will give the same p value as the equal-variances two-sample t test of

$$H_0: \mu_{\text{app}} - \mu_{\text{no app}} = 0 \text{ versus } H_1: \mu_{\text{app}} - \mu_{\text{no app}} \neq 0. \quad \text{Cool!}$$

Exercise: Show that in the above setup we have

$$\frac{\bar{Y}_2 - \bar{Y}_1}{S_{\text{pooled}} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\hat{\beta}_1}{\hat{\sigma} / \sqrt{S_{xx}}}.$$

Do it in steps, showing:

1. $\hat{\beta}_0 = \bar{Y}_1$
2. $\hat{\beta}_1 = \bar{Y}_2 - \bar{Y}_1$
3. $\hat{\sigma} = S_{\text{pooled}}$
4. $1 / \sqrt{S_{xx}} = \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

Appendicitis example (cont)

Prepare the data:

```
Y <- data$hem  
x <- as.numeric(data$diag == "appendicitis")  
head(cbind(Y,x))
```

	Y	x
[1,]	14.8	1
[2,]	15.7	0
[3,]	11.4	0
[4,]	13.6	0
[5,]	12.6	1
[6,]	12.5	0

```
summary(lm(Y~x))
```

$$\mu_1 = \beta_0 + \beta_1$$
$$\mu_2 = \beta_0$$

$$\hat{\beta}_0 = \hat{\mu}_2 = \bar{x}_2$$

$$\hat{\beta}_1 = \hat{\mu}_1 - \hat{\mu}_2 = \bar{x}_1 - \bar{x}_2$$

Call:

```
lm(formula = Y ~ x)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.1323	-0.7323	-0.0323	0.6677	4.1677

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.37316	0.06375	209.782	<2e-16 ***
x	-0.04087	0.08305	-0.492	0.623

Same as the
2-sample t-test
p-value

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.128 on 760 degrees of freedom

Multiple R-squared: 0.0003186, Adjusted R-squared: -0.0009968

F-statistic: 0.2422 on 1 and 760 DF, p-value: 0.6228

Automatic if we designate the predictor as a “factor” (but watch sign!).

```
x <- as.factor(data$diag)
summary(lm(data$hemo ~ x))
```

Call:

```
lm(formula = data$hemo ~ x)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.1323	-0.7323	-0.0323	0.6677	4.1677

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.33229	0.05322	250.490	<2e-16 ***
xno appendicitis	0.04087	0.08305	0.492	0.623

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.128 on 760 degrees of freedom

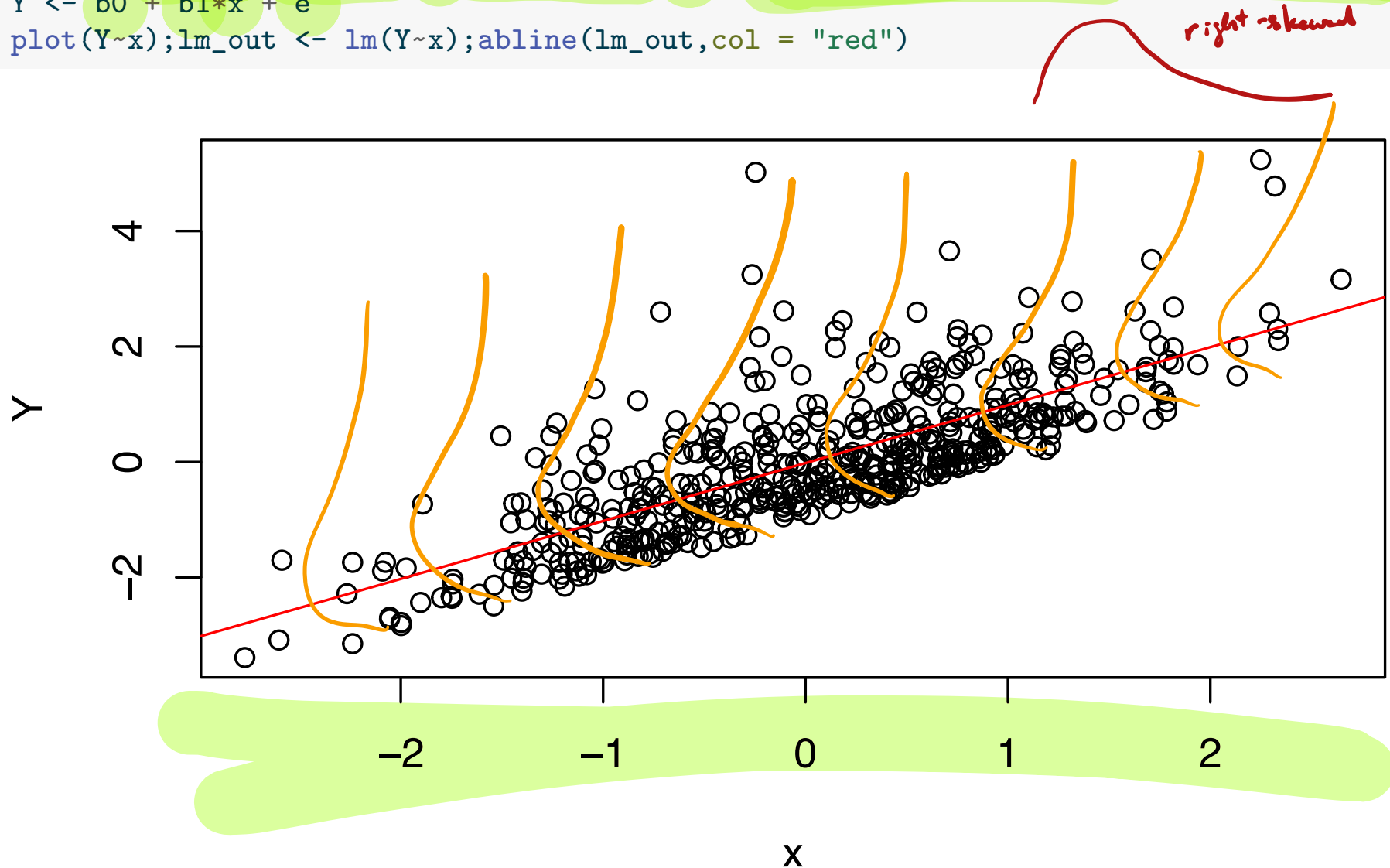
Multiple R-squared: 0.0003186, Adjusted R-squared: -0.0009968

F-statistic: 0.2422 on 1 and 760 DF, p-value: 0.6228

The importance (or unimportance) of Normal error terms

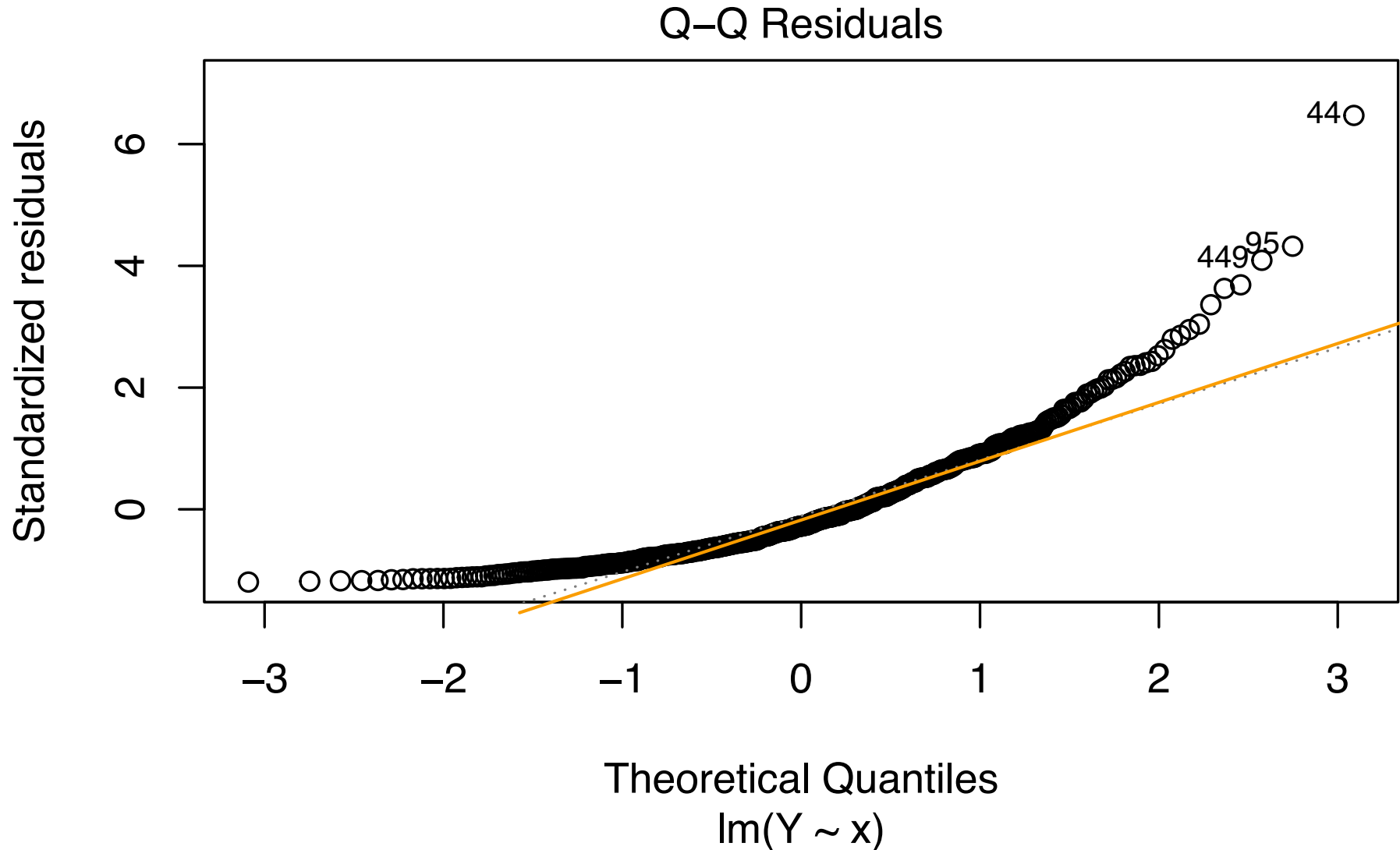
Generate some data with a right-skewed error term distribution:

```
b0 <- 0; b1 <- 1; n <- 500; x <- rnorm(n); e <-  $\varepsilon$ rgamma(n, shape = 3/2, scale = 2/3) - 1  
Y <- b0 + b1*x + e  
plot(Y~x); lm_out <- lm(Y~x); abline(lm_out, col = "red")
```



Check the Normal quantile-quantile plot of the residuals.

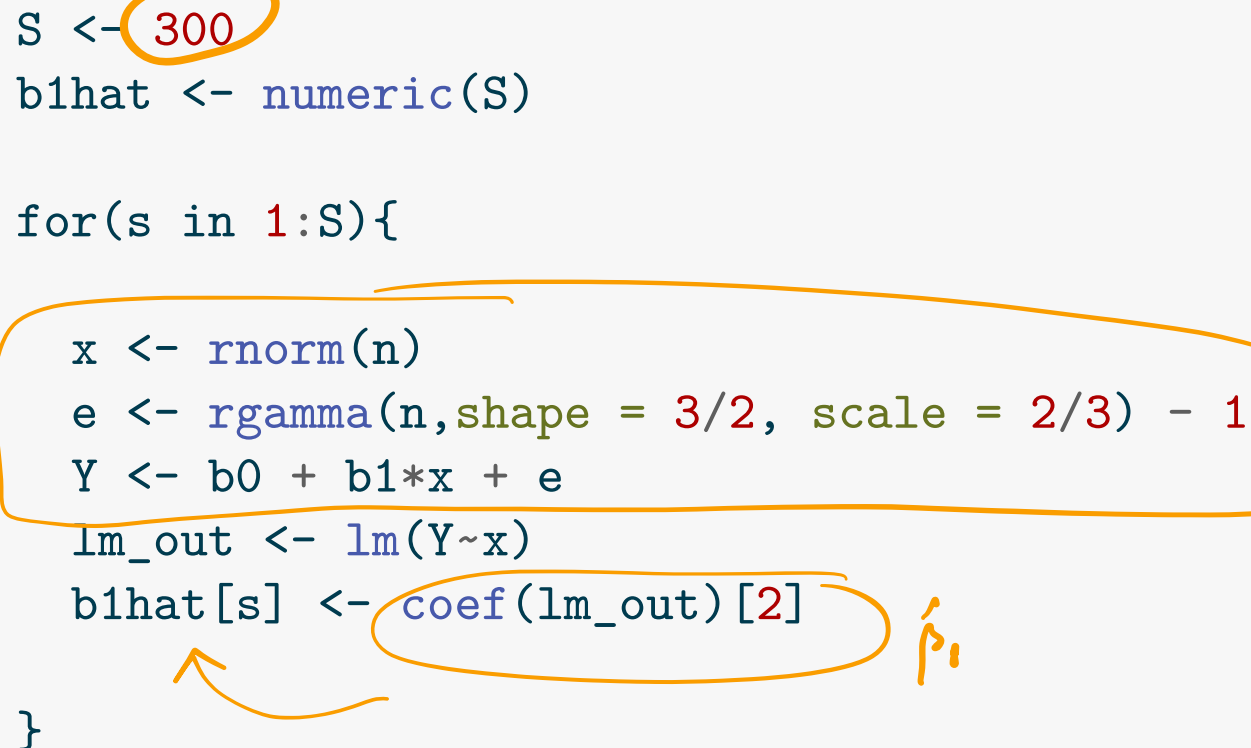
```
plot(lm_out, which = 2)
```



Generate a large number of such data sets and obtain $\hat{\beta}_1$ for each one.

```
S <- 300
b1hat <- numeric(S)

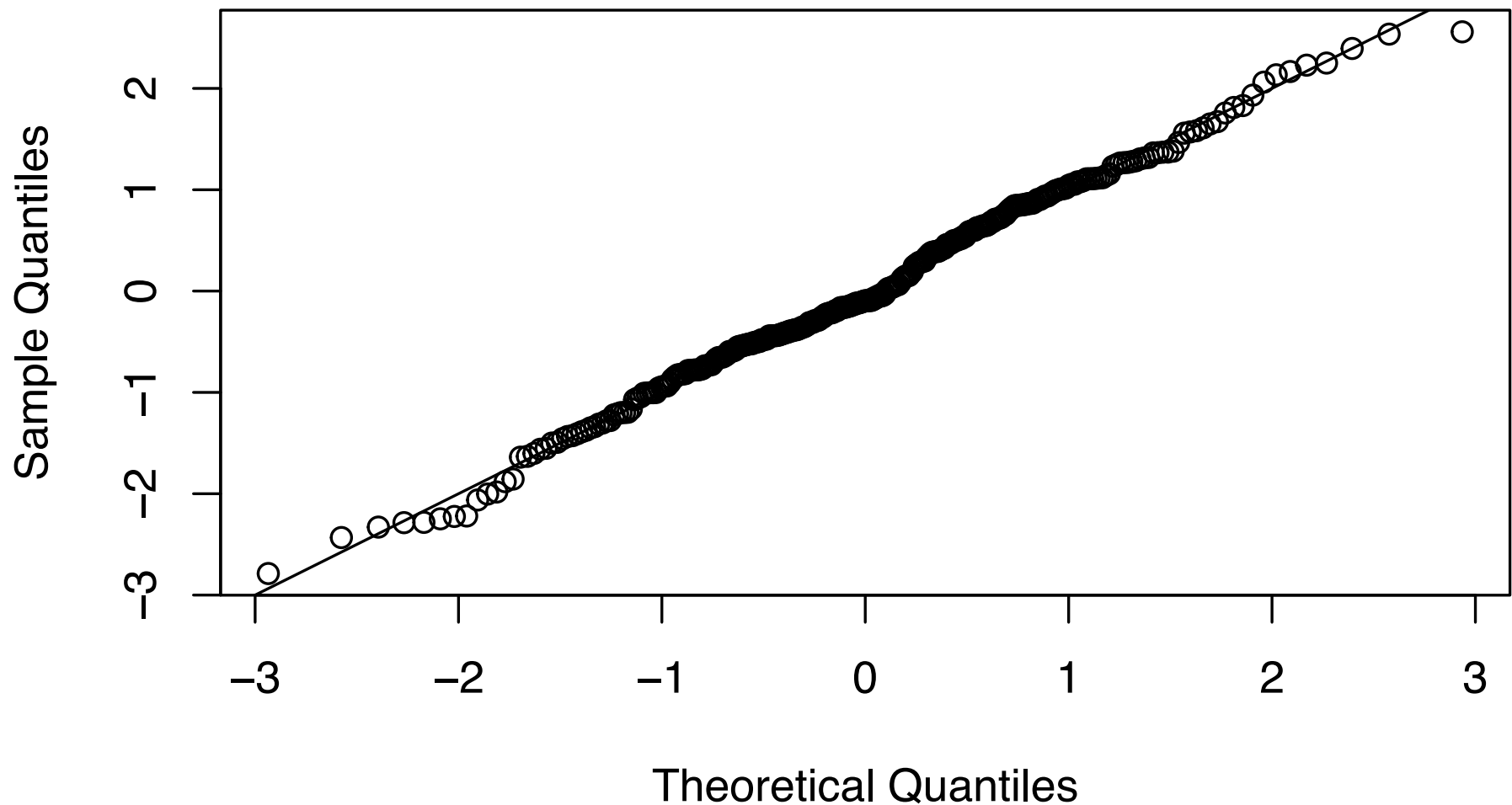
for(s in 1:S){
  x <- rnorm(n)
  e <- rgamma(n, shape = 3/2, scale = 2/3) - 1
  Y <- b0 + b1*x + e
  lm_out <- lm(Y~x)
  b1hat[s] <- coef(lm_out)[2]
}
```



Check if the $\hat{\beta}_1$ values have a Normal distribution.

```
qqnorm(scale(b1hat))  
abline(0,1)
```

Normal Q–Q Plot



References

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